



Estimation and model-based combination of causality networks among large US banks and insurance companies[☆]



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ABSTRACT

Causality is a widely-used concept in theoretical and empirical economics. The recent financial economics literature has used the standard Granger causality to detect for the presence of contemporaneous links among financial institutions, that, in turn, determine a network structure. Subsequent studies have combined the estimated networks with traditional pricing or risk measurement models to improve their fit to empirical data. In this paper, we provide two contributions. First, we show how to use a linear factor model as a device for estimating a combination of several networks that monitor the links across variables from different viewpoints. Second, we highlight the advantages of combining quantile-based methods with the Granger causality when the focus is on risk propagation. The empirical evidence supports our contributions.

1. Introduction

A growing strand of the financial economics literature, related to financial contagion analyses, has investigated the role of links among financial institutions as channels along which shocks spread through the financial system. These links might be read as financial networks. The early literature on financial networks has analysed the transmission channel through which financial contagion spreads within the financial system. The main objective was therefore to identify financial networks by using appropriate criteria to detect links among the involved institutions. Nevertheless, this approach presupposed that such networks were, by definition, unique. Acemoglu et al. (2012) used a network structure based on input–output relationships and showed that aggregate fluctuations may originate from microeconomic shocks to firms. Kelly et al. (2013) showed how stock volatility relates to the customer–supplier connectedness. Billio et al. (2014) used a contingent claim analysis and network measures to underscore the links among sovereign, bank and insurance credit spreads. Moreover, there is a growing number of works proposing competing or alternative approaches for estimating networks among financial institutions, markets, countries and (not necessarily financial) assets. Among them, we refer to Billio et al. (2012), Diebold and Yilmaz (2014, 2015), Hautsch et al. (2012, 2013, 2014, 2015), Barigozzi and Brownlees (2019), Ozdagli and Weber (2017) and Corsi et al. (2015). All these contributions have in common that they refer to either a financial or an economic playground.

Despite the growing interest in the financial network topic, and the increasing understanding that there might be different channels over which financial contagion spreads, the literature on combinations of financial or economic networks is still limited.

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In fact, the different approaches used to estimate networks highlight different potential channels for the spread of both contagion and risk. As a result, they might co-exist within the same financial system. It is then important to allow for the possibility of combining them, obtaining a more complete picture of the risk spreading within the financial system. This paper provides a possible solution building on a multiplex network, that is, a collection of networks (the so-called layers) existing among a set of institutions, that we formally define in Section 2. Ideally, the constituents of a multiplex network represent the outcome of different approaches used to either estimate or identify the links among the analysed companies. The literature on multiplex networks is relatively recent in finance and, to the best of our knowledge, includes only a few papers focusing on contagion as well as on topological properties. Bargigli et al. (2015) analysed the Italian interbank market using a multiplex network, distinguishing between five layers: unsecured overnight, unsecured short-term (up to 12 months), unsecured long-term, secured short-term and secured long-term transactions. Bargigli et al. (2015) also considered an aggregated network, computed by accumulating the available layers. Following the authors, each layer has a different topological property and persistence over time. However, only the overnight market was representative of the aggregate network among all the layers. Léon et al. (2014) studied the Colombian sovereign security market and built a multiplex network with three different layers corresponding to the three Colombian environments trading and registering sovereign securities. They found a strong nonlinear effect for the aggregate network value due to the fact that the principal layer did not transfer its properties to the aggregate network or multiplex. Montagna and Kok (2013) developed a multi-layered interbank network model by using an agent-based setting and focusing on a sample of large EU banks. The authors detected the existence of non-negligible nonlinear effects influencing the shock propagation across individual banks. In other words, when different layers were considered simultaneously, the contagion effect was larger than the sum of the contagion-induced losses observed when these layers were analysed separately. Finally, Molina-Balboa et al. (2014) analysed the multiplex structure of bank exposures within the Mexican banking system. They associate layers with different financial instruments and provide insights on the interdependence among banks, with relevant implications from a systemic risk perspective.

By postulating the existence of a multiplex network, we take a step forward and combine the information contained in the layers, analysing their impact and role within a financial economic model. In particular, we focus on the linear factor model augmented with network dependence introduced by Billio et al. (2017). We show how the model can be generalized to accommodate for the presence of a multiplex network. Furthermore, we provide a model parametrization which offers two useful interpretations of the estimation output. First, the model parameters enable the statistical significance of the information contained in each layer to emerge. This highlights the relevance of the various approaches in measuring and identifying a network within the model, providing an interesting financial interpretation. This analysis might be relevant for policy purposes and market monitoring, as it might identify the most relevant risk-spreading channels. Second, the estimated model parameters reveal a composite network obtained by combining the multiplex network layers. As a result, the model-based combination provides an overall picture of the links existing among the analysed variables, as measured by using different approaches (i.e., those behind the various layers of the network), as well as accounting for their respective relevance.

The model-based combination of networks represents our first important contribution. Our second relevant contribution concerns the estimation of networks. Several authors have already discussed possible ways to obtain a network of financial assets, as we said above. Among the various methods, the Granger causality approach used by Billio et al. (2012) is taken as our starting point. However, when focusing on the risk dimension, we typically focus on variances or, more generally, on systemic and/or systematic risk measures. Nevertheless, Granger causality detects relations in mean. As a result, we do not believe that Granger causality is the most appropriate method to quantify the risk interconnections among financial assets. This prompts us to discuss a set of alternative approaches to estimate financial networks based on causality relationships that go beyond the classical Granger causality. We survey a collection of methods that have a common denominator: they all resort to the quantile regression approach of Koenker and Bassett (1978), in either a parametric or a nonparametric representation. By using these methods, we show how we can identify causality among quantiles of the modelled variables. These quantile-based causality detection methods lead to the estimation of causality networks mimicking the approach of Billio et al. (2012), but oriented towards a risk dimension. This is immediate when focusing on the left quantiles of a random variable. As a further approach, we also show that the nonparametric quantile causality test of Jeong et al. (2012) represents a further alternative to the traditional Granger causality for building causality networks. It is important to highlight that, although the implemented methods focus on the same object, that is, a given r th quantile of the response variable, they differ in capturing the causal impact of the explanatory variable and, therefore, provide a different source of information. In particular, when using the standard quantile regression (Koenker and Bassett, 1978) we detect a linear impact of the explanatory variable using its entire distribution. In contrast, we allow for a nonlinear impact by using the non-parametric causality test (Jeong et al., 2012). Finally, when using the quantile-on-quantile method, we allow for a linear impact on the r th quantile of the response variable (y_r), but focusing on the neighbourhood of the θ th quantile of the explanatory variable (x_r).

We stress that, by focusing on the quantiles of the modelled variables, we have a clear risk interpretation about the estimated networks. Indeed, the left quantiles are closely related to traditional risk measures, such as the Value-at-Risk and the Expected Shortfall. Moreover, these networks will monitor the spreading of shocks or the links among the modelled variables when they are in a high-risk state, and not when they are in a mean (or median) state, as captured by the Granger causality. Our purpose is consistent with two other recent contributions. We refer to Hong et al. (2009) and Corsi et al. (2015), who focus on causality among tail events. The Granger causality in risk in Hong et al. (2009), and also used by Corsi et al. (2015), is based on the occurrence of tail events and detects the possible causality among such events. We might associate the tail with a conditional quantile and estimate it with quantile regressions. We differ from the previous papers in that we focus on the causality within the assessment of the conditional quantiles, not on the occurrence of tail events. We thus consider the causality on the risk measures rather than the causality on the occurrence of extreme events.

We suggest different approaches to build causality networks, which involve monitoring the links across assets with views that might complement the (mean-based) Granger causality. These competing networks represent the input for our model-based network combination. The model also allows us to identify the relevance and the role of the various causality networks. We empirically validate our two main proposals concerning the use of quantile causality to infer the network structure across a set of (financial) variables – and the model-based combination of causality networks. By using three different datasets – US industrial portfolios, banks and insurance companies—and focusing on two different samples, that include, respectively, the periods 2006–2008 (affected by the global financial crisis) and 2011–2015, we provide evidence of the different network structures derived from Granger and quantile causality.

Our results suggest that quantile causality networks are denser than Granger causality networks. This is a relevant finding in terms of systemic risk interpretation. Indeed, a denser network points out a much larger set of links and, thus, a possibly greater systemic effect of shocks. The networks derived from the nonparametric quantile causality test of Jeong et al. (2012) also have a clear core–periphery structure, which might help us to identify the most systemically relevant companies or sectors. Moreover, by resorting to the model-based combination approach, we show how the combined networks have a different relevance across the different datasets. In particular, the quantile causality methods turn out to be more relevant than the standard Granger causality in the resulting composite network. In addition, the linear factor model augmented with a plurality of networks provides residuals that are, on average, less correlated when compared to either the standard linear factor models or the network augmented linear factor model proposed by Billio et al. (2017), which is based on the Granger causality only. Furthermore, the risk factor exposures sensibly change when contemporaneous dependencies across assets are taken into account. Finally, we complement our findings by showing that, when focusing on the risk dimension, a simple long-short portfolio strategy building on the combined network centrality outperforms similar strategies based on a single network.

The paper proceeds as follows. Section 2 discusses the approach we propose to obtain a model-based composite network, extending the model of Billio et al. (2017). Section 3 provides the empirical findings, focusing on both the estimation of quantile causality networks and the model-based combination of networks. Section 3 also includes a set of robustness checks. Section 4 concludes. A web appendix accompanies the paper; it includes additional tables and graphs, along with further methodological notes.

2. Model-based network combination

In this section, we introduce how we obtain a model-based combination of a set of different networks. We assume that all the networks are available and thus we do not estimate the structure of the various networks, but focus only on their optimal combination. We start by formally defining both a simple and a multiplex or multilayer network.

A network (or graph) $G = (V, E)$ is a collection of vertices (or nodes) V and edges (or links) E , where $E \subseteq (V \times v)$, typically represented by using a $K \times K$ adjacency matrix W , where K is the number of nodes. In particular, W is a binary matrix in which each element $w_{i,j}$ can take only the values of 0 and 1. When $w_{i,j} = 1$, node i is linked to node j , whereas $w_{i,j} = 0$ points out the absence of such relation. We set $w_{i,i} = 0$, as we exclude self-loops, for $i = 1, \dots, K$. When W is symmetric, the corresponding network is defined as *undirected*. In this case, the direction of links is not relevant, as they represent mutual relationships between a pair of nodes. In contrast, when W is not symmetric, the corresponding network is defined as *directed*. In this case, $w_{i,j} = 1$ points out an influence of node i on node j , whereas we do not observe any impact from i to j when $w_{i,j} = 0$. Note that $w_{i,j}$ is not necessarily equal to $w_{j,i}$ in *directed* networks, for $i \neq j$. When focusing on *undirected* networks, we can compute, for each node, two different measures, defined, respectively, as *outdegree* – i.e., the number of outgoing links starting from a specific node – and *indegree*—i.e., the sum of the ingoing links on the node of interest. We can also relax the assumption that $w_{i,j}$ can take only two possible values. Indeed, we can allow $w_{i,j}$ to take values in the (positive) real axis. In this case, a network is defined as *weighted* (as opposed to *unweighted*) because its links have also a *size/relevance*. As a result, each edge simply represents the presence of a connection in unweighted networks, whereas, in weighted networks, edges also contain information about the strength of the relations among nodes.

It is worth noting the parallelism between the adjacency matrix and the spatial (proximity) matrix used in spatial econometrics. The latter describes how the subjects of an analysis are spatially related. In such a different framework, the distance between subjects is (generally) measured on a physical (natural) playground and connections correspond to neighbourhood relationships. Therefore, if the element $w_{i,j}$ is equal to 1, the element j is a neighbour of the element i , following the spatial econometric nomenclature. The spatial matrices are symmetric in standard applications, as neighbouring relationships are symmetric. It is also common practice to focus on proximity matrices that are row-normalized. These matrices can then be interpreted as directed weighted networks, as symmetrical relations might become asymmetrical after normalization.

Within a financial framework, networks are generally directed and weighted when they are measured by means of balance sheet quantities. For instance, Billio et al. (2016) used BIS cross-holdings, Diebold and Yilmaz (2014) developed an approach based on variance decompositions of target series, Acemoglu et al. (2012) used the Input–Output matrix. A network might also be directed and unweighted. Relevant examples are the networks obtained using Granger causality (Billio et al., 2012) or Granger Causality in the tails (Corsi et al., 2015). A network might even be unweighted and undirected, as in the economic sector case of Caporin and Paruolo (2015).

In the following, we assume that the strength of the edges is normalized at each node, as in a row normalization of the adjacency matrix W . We also assume that we are dealing with directed networks (the most general case that we could consider). A multiplex (or multilayer) network is a collection of d networks organized into d different planes, or layers. In principle, the number of nodes is not necessarily the same across networks, so that the total number of nodes in a multilayer network is equal to $\sum_{i=1}^d V_i$, where

V_i is the number of nodes or vertices in the i th layer. Further, if the graphs (associated with the networks) are simple (there are no loops), the maximum number of edges (connections between nodes) equals $\prod_{i=1}^d \frac{V_i(V_i-1)}{2}$, since connections across layers are also allowed. In a financial framework, a multilayer network has an even simpler structure. Indeed, the nodes are the same across layers because we have measures of connections on the same financial institutions, but we use different approaches to estimate the networks. We also assume that there are no connections among layers, leaving us a collection of networks that are unrelated, but all referring to the same nodes.

This work aims to combine financial networks by building on a variant of a financial economic model, that is, the linear factor model augmented with a spatial link, following the approach of [Billio et al. \(2017\)](#). The advantage of our model lies in that the combination of the various layers of the multiplex network is weighted with a set of coefficients appearing in the model parametrization. As well as representing the elements for a convex linear combination of layers, the coefficients would then enable the significance and relevance of each layer to be assessed. We briefly review the model proposed by [Billio et al. \(2017\)](#) in the next subsection. Here, we also introduce the model generalization in the presence of a multilayer network. We then discuss the approach used to derive a composite network from the model estimates.

2.1. A linear factor model with multilayer spatial dependence

To obtain a composite network, we generalize the model of [Billio et al. \(2017\)](#), a *revisited* multifactor model capable of taking into account the contemporaneous links among a set of assets (as measured by a network) together with the presence of common factors. In this model, the returns equation is defined as follows:

$$A(R_t - \mathbb{E}[R_t]) = \bar{\beta}F_t + \eta_t, \quad (1)$$

where A is the matrix of the contemporaneous relationships among the endogenous variables – i.e., the returns – coexisting with the exposure to a set of common factors (F_t); R_t is the vector of asset returns observed at time t , whereas their expected values are included in $\mathbb{E}[R_t]$; $\bar{\beta}$ includes the structural asset exposures to the common factors, whereas η_t is the vector of the structural idiosyncratic residuals, which have a diagonal covariance matrix denoted as Σ_η .

The standard multifactor model can be seen as the reduced form of (1). Under the assumption that A is invertible, it has the following reduced-form representation:

$$\begin{aligned} R_t &= \mathbb{E}[R_t] + A^{-1}\bar{\beta}F_t + A^{-1}\eta_t \\ &= \mathbb{E}[R_t] + \beta^*F_t + \epsilon_t^*, \end{aligned} \quad (2)$$

Starting from the reduced form, and in the presence of contemporaneous links across assets, [Billio et al. \(2017\)](#) make the point that: (i) the reduced-form residuals $\epsilon_t^* = A^{-1}\eta_t$ are cross-correlated; and (ii) the reduced-form betas, $\beta^* = A^{-1}\bar{\beta}$, are a nonlinear function of the structural betas and of the contemporaneous relation matrix A . The intuition of [Billio et al. \(2017\)](#) is to provide a structure for A driven by the existence of a network representing the (contemporaneous) links across the assets. Therefore, they propose to make the matrix A an affine function of a network, $A = (I - \rho W)$, where ρ is a scalar parameter indicating the spatial dependence of the returns on the network, whereas W is an exogenous spatial matrix, or the adjacency matrix of a network. Such a choice also enables us to cope with the identification issues of the simultaneous equation system in (1). Following [Anselin \(1988\)](#) and [LeSage and Pace \(2009\)](#), the model includes a spatial autoregressive component. The estimation of the spatial parameter ρ follows from concentrated likelihood methods. [Billio et al. \(2017\)](#) also allow for a more flexible parametrization of A . In particular, they consider two relevant extensions:

- the use of asset-specific impacts from the network, which corresponds to a contemporaneous parameter matrix A with equation-specific network impacts, namely $A = I - \mathcal{R}W$, where $\mathcal{R} = \text{diag}(\rho_1, \rho_2, \dots, \rho_K)$ is a diagonal matrix, whereas the coefficient ρ_i represents the impact of the network on each asset; and
- the possibility that the spatial matrix, or network, evolves over time, thus leading to W_t . This is particularly relevant in finance because W is not related to a *physical* distance between subjects, but measures proximity or connections starting from variables liable to temporal differences.

In the following, for the sake of simplicity, we assume the networks are time-invariant, though the approach that we introduce can be easily extended to the case of dynamic networks. In the presence of a multilayer network, we generalize the model of [Billio et al. \(2017\)](#) to accommodate for the presence of several adjacency matrices. In the spatial econometrics literature, we have some examples associated with the existence of richer spatial dynamics. In particular, [Brandsma and Ketellapper \(1979\)](#) propose a higher-order spatial autoregressive model, introducing two spatial matrices and, consequently, two spatial autocorrelation parameters: the first matrix based on the first-order neighbours, the second derived from the higher-order neighbouring relations. In the framework of [Billio et al. \(2017\)](#), the matrix of simultaneous relations A , in the presence of a large number of networks, is easily generalized to:

$$A = I - \sum_{j=1}^d \rho_j W_j, \quad (3)$$

where ρ_j is a scalar coefficient that captures the impact of the network W_j on the contemporaneous links across the assets.

Such a parametrization would give rise to a common impact of each network on the whole collection of modelled asset returns. Therefore, we propose to combine the existence of many networks with a restricted parametrization, accommodating for the presence of various networks, while also allowing for heterogeneous impacts of the networks on the assets. In detail, we suggest the following model specification in the presence of d different networks (i.e., of a multiplex network with d layers):

$$\left[I - \mathcal{R} \left(\sum_{j=1}^d \delta_j W_j \right) \right] R_t = A R_t = \mathbb{E}[R_t] + \beta F_t + \eta_t, \quad (4)$$

where d is the number of layers or networks, $\mathcal{R}$ is a diagonal parameter matrix determining the heterogeneous impact of the networks on the asset returns, whereas δ_j controls the impact and the role of the j th layer in the multiplex network. We remind that the β parameters are unconstrained and that the covariance of the innovations, Σ_η , is diagonal.

The coexistence of many networks in the augmented linear factor model of Billio et al. (2017) has consequences on the identification of the model parameters. To achieve identification, we make the following assumptions:

Assumption A1. $W_j \neq 0$, $j = 1, 2, \dots, d$;

Assumption A2. $W_i \neq W_j$, $i, j = 1, 2, \dots, d$, $i \neq j$;

Assumption A3. The δ_j parameters satisfy the following constraints:

$$\sum_{j=1}^d \delta_j = 1, \quad \delta_j \geq 0, j = 1, 2, \dots, d; \quad (5)$$

Assumption A4. $\text{rank}(W_{i,\cdot}) > 0$, where $\text{rank}(\mathcal{A})$ is the rank of the matrix $\mathcal{A}$, $W_{i,\cdot} = [W_1' |_{\cdot,i} \quad W_2' |_{\cdot,i} \quad \dots \quad W_d' |_{\cdot,i}]$, where $W_j' |_{\cdot,i}$ denotes the i th row of W_j .

Assumption A1 simply rules out the case of absence of edges among a collection of nodes, whereas **Assumption A2** excludes the case where two networks, measured by two different approaches, are identical. If the latter case occurs, we lose identification, as the two networks are indistinguishable, and one of the two must be excluded from the analysis. In contrast, **Assumption A3** requires positivity of the coefficients. This is important for interpretation purposes, whereas the unit sum is needed to avoid that two sets of parameters in $\mathcal{R}$ and δ_j lead to the same composite network, for $j = 1, 2, \dots, d$. The first three assumptions are relevant for the identification of δ_j , whereas **Assumption A4** refers to the parameters in $\mathcal{R}$, ensuring that we do have at least one non-null element in each row of the following combined network:

$$W^* = \sum_{j=1}^d \delta_j W_j. \quad (6)$$

Indeed, the parameter $\rho_{i,\cdot}$ of the diagonal matrix $\mathcal{R}$ is identified if the i th row of W^* has at least one non-null element. Note that, under the previous assumptions and given the structure we impose on $\mathcal{R}$ and Σ_η , the model is overidentified. This leaves space for any richer parametrization in (4), with heterogeneous impacts on returns from groups of layers, for instance. We suggest to estimate the model using the concentrated maximum likelihood method, according to the standard practice in spatial econometrics. If matrix A is known, the factor loadings β and the covariance matrix of innovations Σ_η can be obtained with traditional least squares estimators. We can then concentrate out the latter parameters and use concentrated maximum likelihood to estimate the parameters in A . The maximization must account for the constraints that we impose on δ_j .

The model includes a sort of spatial autoregression, the structure of which is driven by several spatial proximity matrices. A standard requirement for estimations of spatial econometric models is the invertibility of matrix A . When the number of networks is restricted to one ($d = 1$), the non-singularity of A^{-1} is ensured when $\lambda_{\min}^{-1} < \rho < \lambda_{\max}^{-1}$, where $\lambda_{\max}$ and $\lambda_{\min}$ are, respectively, the maximum and the minimum eigenvalues of the spatial/adacency matrix W . In the presence of numerous proximity matrices, the non-singularity of A might be addressed with analytical tools, but only in specific cases. For instance, Lee and Liu (2010) and Elhorst et al. (2012) discuss the case of two matrices W , referring to the estimation of a second-order spatial autoregressive model. These results are not easy to extend to our specification. However, we control for the invertibility of A within the estimation step.

The model offers two important insights. The first, and most relevant, concerns the feasibility of interpreting the summation of the adjacency matrices, W^* , as a new adjacency matrix representing a composite network. This is a direct consequence of the constraints on the coefficients leading to a convex linear combination of adjacency matrices, which are all row-normalized. Therefore, the composite matrix will also be a row-normalized matrix representing a combination of a collection of networks. This combination will be also directed and weighted, as the original networks are directed and weighted. δ_j quantifies the relevance of the j th layer in the resulting composite network, for $j = 1, \dots, d$. As a result, the model generates a composite network, providing, at the same time, insights about the impact of each approach used to build the different layers. Second, the parameters in $\mathcal{R}$ reproduce the heterogeneous reaction of the assets to the composite network. This makes the multiple-network model comparable, in terms of its economic interpretation, to the network of Billio et al. (2017), but with the important difference that its underlying network is composite.

2.2. Causality testing and causality networks

The concept of causality and the tools used to test for its existence date back to the seminal contributions of Granger (1969, 1980, 1988). The purpose of his original contribution was to identify causal relationships in mean across economic variables, and his work attracted considerable interest in the econometrics literature; see, e.g., Geweke (1984) and Hoover (2001). This method was later extended to the analysis of causality in variance. Granger et al. (1986) provided a first definition of causality in variance, whereas Comte and Lieberman (2000) extended and generalized this concept. On the testing side, Cheung and Ng (1996) proposed a test based on cross-correlations, Hafner (2003) introduced a procedure based on the likelihood of competing models, whereas Hafner and Herwartz (2008) presented a Wald-type approach. More recently, contributions on causality between two random variables, that go beyond the first- and second-order moments, have been proposed. In particular, we refer to causality in quantiles – see, e.g., the nonparametric test of Jeong et al. (2012) and the work of Lee and Yang (2014) – and in distribution—see, e.g., Candelon and Topkavi (2016).

The use of Granger causality in estimating financial networks stems from the work done by Billio et al. (2012). The authors associated the statistical evidence of Granger causality in mean with the existence of an edge connecting two assets (or nodes). In a context of market turmoil or when the focus is on systemic risk spreading after local or specific shocks, the standard causality in mean might not be enough, and should be combined with a causality testing that covers the risk dimension. Appendix A provides a brief review of the methods we use in the empirical analysis. First, we describe the approach of Billio et al. (2012), based on Granger causality. We then move to causality in quantiles, focusing on three different approaches: (i) a Granger-type causality building on the baseline quantile regression introduced by Koenker and Bassett (1978); (ii) a generalization of the baseline quantile regression, accounting also for the location of the conditioning (or explanatory) variable over its support, defined by Sim and Zhou (2015) as quantile-on-quantile; and (iii) the nonparametric quantile causality test introduced by Jeong et al. (2012).

As discussed in the introduction, the lower quantiles of stock returns acquire a more relevant role within a financial framework, as they are interpreted as Values-at-Risk. The availability of a number of competing methods for estimating causality networks enables the construction of a multiplex network, in which each layer is associated with a specific network. The multiplex network based on causality links can then be used for the model-based construction of composite networks, as discussed in the previous section. The same criteria apply to the estimation of a single causality network, irrespective of the approach used to identify causality links. It is also important to note that we estimate a causality network starting from historical information on K assets, generally focusing on their returns. As a result, the K assets represent the network nodes. Our purpose is to identify edges connecting each pair of nodes. If all assets were connected, when excluding self-loops, we would have a total number of $K^2 - K$ edges in which case the associated adjacency matrix is symmetric if the network is unweighted.

In all the cases we discuss below, for any pair of assets i and j , the existence of an edge connecting the nodes i and j with a specific direction (say, from i to j) is established from a statistical test. The null hypothesis of the various tests corresponds to the absence of a causality link, while the null hypothesis is rejected in the presence of some form of causality. The type of causality detected clearly depends on the hypothesis tested. For a given test statistic $\mathcal{M}$, and for any pair of assets i and j , we define the edge from i to j as follows:

$$w_{i,j} = \begin{cases} 0 & \mathcal{M} \text{ null hypothesis rejected} \\ 1 & \mathcal{M} \text{ null hypothesis accepted} \end{cases} \quad (7)$$

where $w_{i,j}$ is the element in position i, j within an adjacency matrix.

Given the use of the test statistic $\mathcal{M}$, we denote the corresponding adjacency matrix as $W_{\mathcal{M}}$. The outcome of the test statistic, either rejecting or accepting the null hypothesis, thus identifies the edges of a directed and unweighted network. We do not discuss here a possible generalization for a more flexible network construction, for instance, taking into account also the strength of the causal relations. Such an extension is clearly possible but is left to future research. It should be noted that the networks must be row-normalized before being used in the model presented in Section 2.1.

3. Empirical analysis

3.1. Data description

We conduct the empirical analysis on three different datasets. The first includes the 48 industry portfolios provided by the Kenneth R. French. The second and the third include, respectively, the 25 US banks and the 25 US insurance companies with the largest market value. Appendix B reports the list of the companies and industry portfolios we consider.¹ Using the three datasets, we can test for the presence of causality links from a general economic standpoint, or by looking at the companies that were mostly affected by the recent financial crisis.

For the three datasets, we use the daily returns observed in two different samples. The first spans the open-market business days between 3 January 2006 and 31 December 2008. The second covers the period between 3 January 2011 and 31 December 2015. The former is characterized by 755 daily returns for each time series, whereas the latter records 1258 observations for each industry portfolio or firm. We find relevant differences between the two samples. Indeed, the first includes the years of the global

¹ The series of the industry portfolios are available at <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/>. The dataset including the banks and the insurance companies has been obtained from Thomson Reuters Datastream.

financial crisis, whereas the second covers a period of upward trend in the financial market, excluding the initial recovery period after the crisis. Our purpose is to test whether there is a different dependence structure among the variables during periods of market distress vis-à-vis periods with a positive market trend. We proceed with our estimation of causality networks by focusing on the daily returns.

Focusing on the first period, we select the 25 US banks and the 25 US insurance companies with the largest market value at the beginning of 2006. A similar criterion is used for the second period, taking into account the companies' market value at the beginning of 2011. Appendix B reports a list of the selected banks and insurance companies for both samples. We stress that our objective is to obtain an overview of the differences in the links across nodes (banks, insurance companies or economic sectors) when changing the approach used to ascertain their existence. In doing so, we switch from Granger to quantile-based causality, to obtain the optimal combination of the competing available networks.

Tables C.1–C.2 in Appendix C report the main descriptive statistics computed for the banks, for the periods 2006–2008 and 2011–2015, respectively. Likewise, the statistics computed for the insurance companies (48 industry portfolios) are reported in Tables C.3–C.4 (C.5–C.6). In all cases, the average returns are lower during the period 2006–2008. This was expected, due to the effects of the global financial crisis. The interval 2006–2008 is also characterized by a greater uncertainty, as quantified by the standard deviation and the interquartile range. The greater risk during the period 2006–2008 is also evident when considering the minimum and the maximum returns. The distributions of these returns are fat-tailed (especially in the first period), consistent with the stylized facts highlighted by Cont (2001). The distributions are also skewed. Finally, we use the Ljung–Box test for mean serial dependence on both the returns and the squared returns, using 1, 5 and 10 lags. The outcome shows no serial dependence on returns, while squared returns (a proxy of variances) are highly correlated, suggesting the presence of heteroskedasticity. In short, our descriptive analysis shows significant differences between the periods 2006–2008 and 2011–2015. These differences are a clear consequence of the turmoil affecting the US economy during the former period. We now take a closer look at the dependencies among companies or economic sectors by using causality analyses.

3.2. Granger and quantile causality networks

In building the causality networks, we extend the existing literature by combining the standard Granger testing approach with methods based on dependencies across quantiles, as described in Appendix A.1. The methods based on quantile causality depend on the quantile at which the analysis is performed. Therefore, we must first specify which quantiles we use to conduct the dependence or causality analysis. Our analysis illustrates the changes occurring in the causality networks when switching from Granger to quantile causality, without identifying the optimal quantile level over which to measure quantile causality. We therefore decided to fix the quantiles of interest a priori. In particular, we chose three quantile levels: 0.1, 0.5 and 0.9, corresponding to the tails and to the centre of the returns distributions. Note that the approach of Sim and Zhou (2015) depends on two quantiles: the τ th (for the response variable) and the θ th (for the explanatory variable) quantiles. We set both quantiles at the same level, that is, $\theta = \tau$ (see Appendix A for details on the quantile causality methods we consider).

When evaluating the quantile regression parameters, we estimate their standard errors by using the xy -pair bootstrap method (Efron and Tibshirani, 1993) with 5000 replications. This approach provides accurate results without assuming any particular distribution for the error term. When a kernel function is involved (see Appendix A for details), we use the Gaussian one. Following Balcilar et al. (2017) and Bonaccolto et al. (2018), we estimate the optimal bandwidths involved in the (Jeong et al., 2012) test using the least squares cross-validation method of Rudemo (1982) and Bowman (1984). We also need to specify the lag orders for both the dependent and the explanatory variables. To limit the computational burden, we fix a priori both lag orders to be 1. Finally, as for the quantile-on-quantile method, the objective functions depend also on a bandwidth (see Appendix A for details). Larger values of the bandwidth make smaller the variance of the estimates, but, on the other hand, would increase their bias. The opposite holds when h takes lower values. Optimizing the value of h by using ad-hoc statistical methods is still an open issue in a quantile-on-quantile framework; we do not analyse this aspect in the present study and leave it for future research. In contrast, we follow Sim and Zhou (2015) and use a plug-in bandwidth equal to 0.1. We remind that the bandwidth applies to a quantile level. Our choice allows avoiding estimating the parameters with a small number of data points (as it is the case with smaller bandwidths) and, at the same time, to keep the focus around the quantiles of interest. We set the confidence level at 5% for judging the significance of the coefficients and testing the null hypothesis behind the tests for Granger or quantile causality. To control for the possible impact of the common dependence on the market, we test for Granger causality as well as for the various forms of quantile causality, also including the supercomposite US market index within the corresponding models.²

In the remainder of this paper, we identify the various networks that we estimate by means of acronyms. For networks estimated using quantile regression approaches (the baseline case, the quantile-on-quantile and the nonparametric test), each acronym consists of four or five characters. The first and the second ones identify the method: QB for baseline quantile regression, Qo for quantile-on-quantile and QN for nonparametric quantile. The third and fourth characters identify the quantile. For instance, QB10 refers to the network estimated by using the baseline quantile regression at the 10% quantile level. As for the Qo method, the reported quantile is used for both the dependent and the explanatory variables. We use GR for the standard Granger causality networks. Finally, we add a final letter F to the acronym when we use a common factor in the estimation of the causality networks.

Fig. 1 displays four different networks for the largest US banks: the Granger causality and the three different quantile causality networks estimated focusing on the 10% quantile level. All the networks refer to the financial crisis period (period 2006–2008)

² We obtain the market return from the Fama/French 3 Factors dataset provided by Kenneth R. French and available at <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/>.

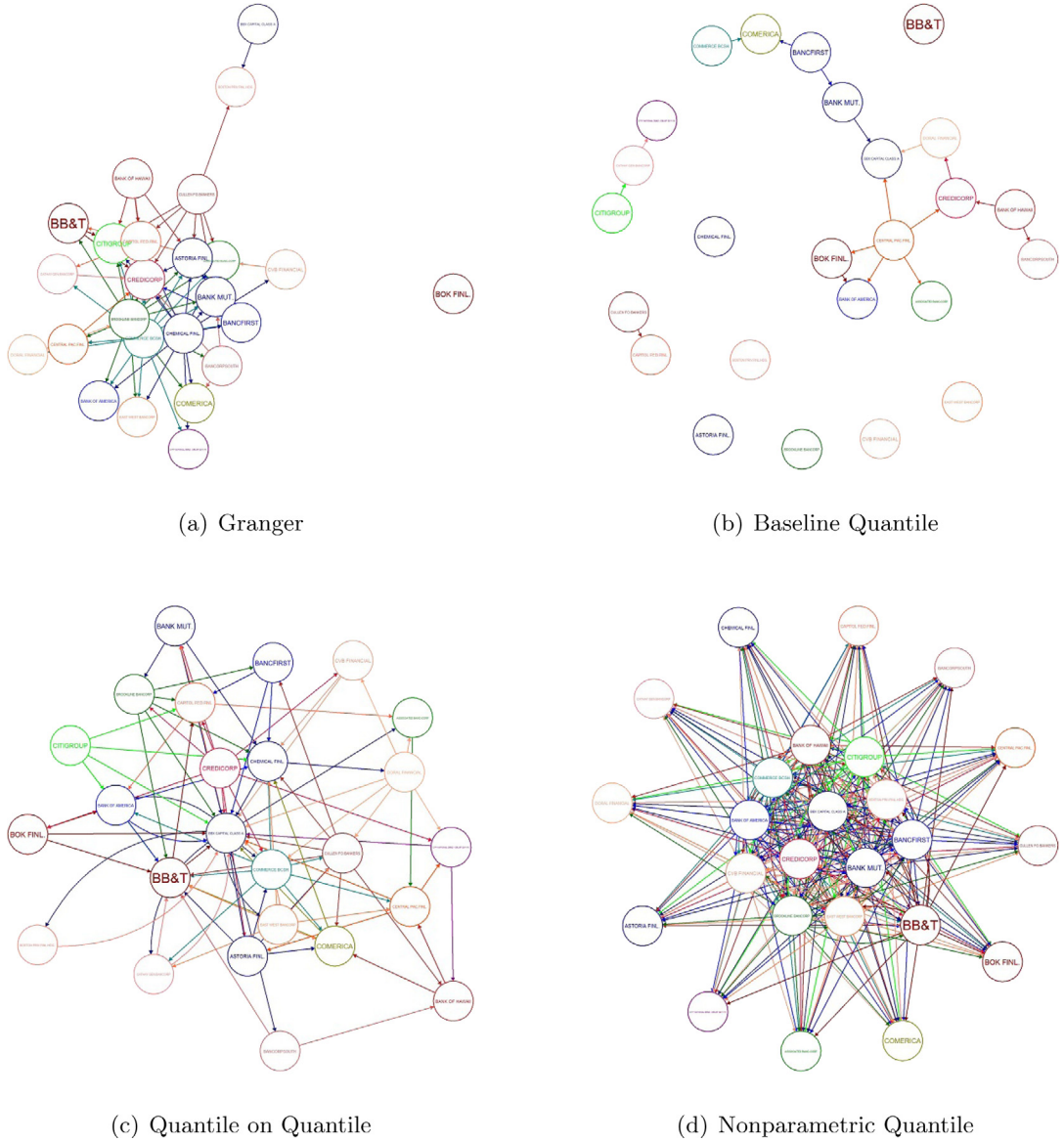


Fig. 1. The figure displays 4 different networks for the period 2006–2008 focusing on the first 25 banks ordered by market capitalization and listed in Appendix B. Panel (a) reports the network derived from the standard Granger causality approach of [Billio et al. \(2012\)](#). Panels (b), (c) and (d) report the networks obtained by using the baseline quantile regression, the quantile-on-quantile approach and the nonparametric quantile causality methodology, respectively, at the 10% quantile level. All the estimates exclude the presence of common factors. We use the Fruchterman & Reingold algorithm for network visualization.

and are estimated without any common factor. Appendix D includes graphs for all the estimated causality networks. Here, we report results for the 10% quantile level because we wish to shed some light on the possible differences between Granger causality (which focuses on the mean) and quantile causality (which places more emphasis on the tails) when the involved companies (or industry portfolios) are in distress. Networks building on either the median or the 90% quantile level causality are available in Appendix D. [Fig. 1](#) shows that the baseline quantile regression (QB) produces a network characterized by a relevant set of isolated nodes. In contrast, the network derived from the nonparametric quantile causality (QN) has a greater density. When comparing the different datasets, we observe that banks are less connected whatever method is used (see Appendix D). The nonparametric quantile causality test also generates networks characterized by a similar topology for all institutions and portfolios; in particular, it is easy to distinguish between a kernel and a periphery.³ We compare in [Fig. 2](#) the networks estimated for the periods 2006–2008 and 2011–2015, by using the banks dataset and two specific causality methods: the Granger and the nonparametric quantile causality (10% quantile level). We can see some changes in the Granger network structure when moving from the earlier to the later period.

³ We use the Fruchterman & Reingold algorithm for network visualization, [Fruchterman and Reingold \(1991\)](#).

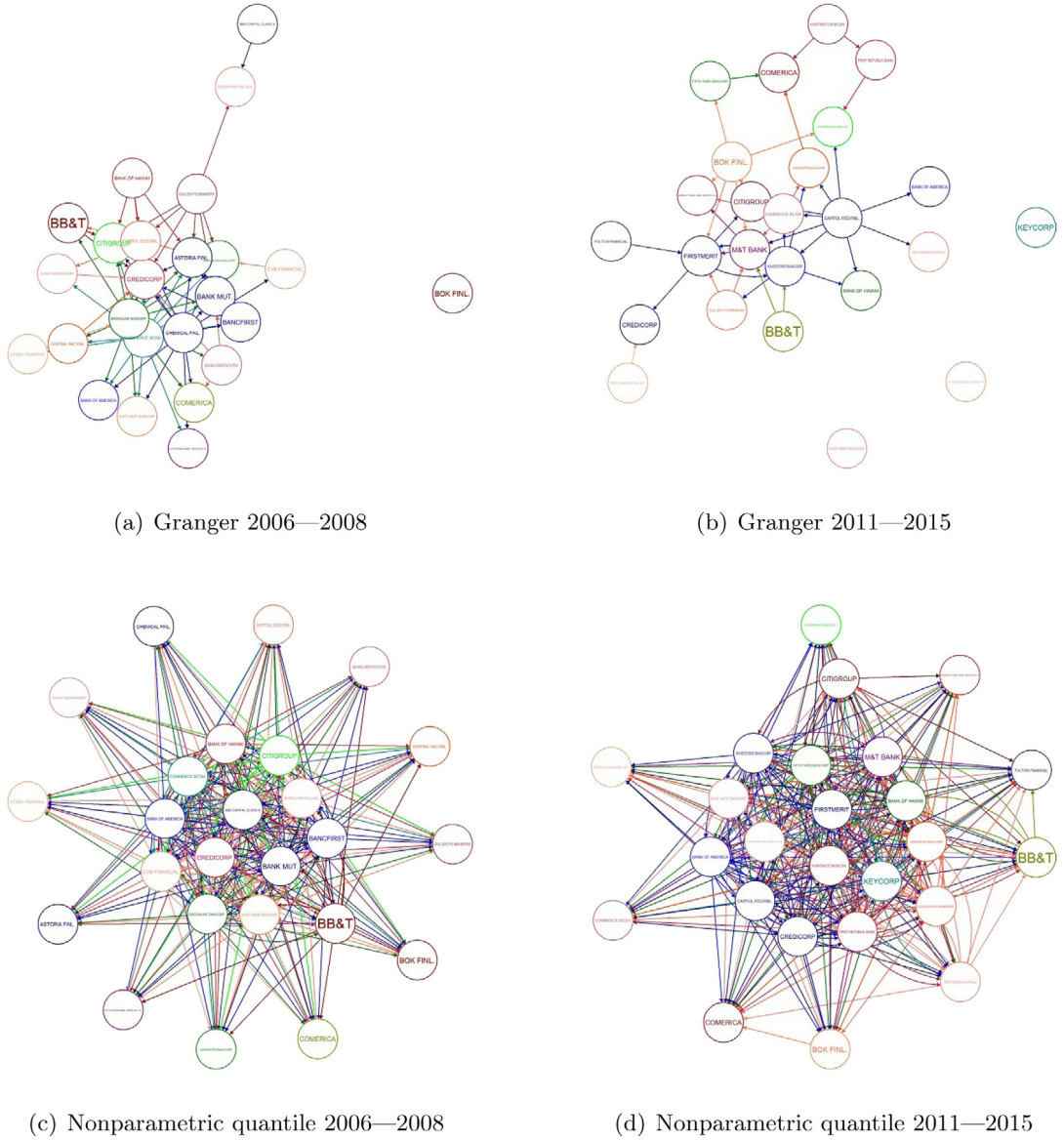


Fig. 2. The figure displays the network of the 25 banks ordered by market capitalization (see Appendix B) for different methods and samples. Panels (a) and (b) report the networks obtained by using the standard Granger causality method for the 2006–2008 and 2011–2015 samples, respectively. Panels (c) and (d) report the networks obtained by using the nonparametric quantile causality test at the 10% quantile level for samples 2006–2008 and 2011–2015, respectively. All the estimates exclude the presence of common factors. We use the Fruchterman & Reingold algorithm for network visualization.

These changes are less clear when we consider the nonparametric quantile causality, in which case the network still presents a core–periphery structure. This preliminary graphical analysis suggests that there are potentially relevant differences between the networks estimated by using Granger causality and quantile-based causality approaches. The latter have a stronger focus on risk than the former.

Before moving to the combination of the competing approaches (QB, Qo, QN and GR), we compare the individual networks by using summary measures. We consider four indicators: *density*, *assortativity* and two different versions of *eigenvector centrality*. Density monitors the number of connections between nodes. A greater density points out a large number of connections among nodes, which are then more closely related to one another. Assortativity captures the tendency of a node to connect with other nodes having similar properties; it can take values in the range of $(-1, 1)$. Values close to 1 point out that the network has an assortative behaviour, with nodes being connected to other nodes that have similar degrees. Values nearing zero point out that a network becomes similar to a random graph. A disassortative behaviour, when high-degree nodes point to low-degree nodes, corresponds to

Table 1

The table reports the density, the assortativity and the eigenvector centrality of the networks estimated from the following causality methodologies: baseline quantile (QB), quantile—on—quantile (Qo), nonparametric quantile causality (QN) and Granger causality (GR). The second column reports the reference quantile level used to estimate the quantile—based causality networks. The third column indicates whether the market index has been used (Y) or not (N) as a common factor in the estimation of the causality networks. We report the results obtained for the banks dataset in the periods 2006–2008 and 2011–2015.

Model	Quantile	Factor	Density		Assortativity		Eig. Centrality	
			06–08	11–15	06–08	11–15	06–08	11–15
GR	–	N	0.12	0.07	–0.37	–0.22	0.12	0.26
QB	10%	N	0.03	0.04	0.05	–0.13	–	–
QB	50%	N	0.05	0.07	0.03	–0.06	0.16	–
QB	90%	N	0.12	0.13	–0.44	–0.22	0.25	0.11
Qo	10%	N	0.16	0.36	–0.37	–0.28	0.23	0.45
Qo	50%	N	0.06	0.03	–0.43	0	0.34	–
Qo	90%	N	0.17	0.35	–0.53	–0.07	0.25	0.44
QN	10%	N	0.51	0.63	–0.11	–0.2	0.93	0.83
QN	50%	N	0.29	0.33	–0.3	–0.28	0.75	0.72
QN	90%	N	0.57	0.52	–0.15	–0.13	0.9	0.83
GR	–	Y	0.07	0.07	–0.01	–0.24	0.22	0.31
QB	10%	Y	0.04	0.04	–0.26	0.13	0.08	–
QB	50%	Y	0.04	0.09	–0.23	–0.12	0.14	0.29
QB	90%	Y	0.1	0.09	–0.37	–0.34	0.21	0.26
Qo	10%	Y	0.22	0.5	–0.29	–0.24	0.24	0.5
Qo	50%	Y	0.06	0.03	–0.07	–0.25	0.18	–
Qo	90%	Y	0.1	0.51	–0.5	–0.2	0.18	0.57
QN	10%	Y	0.52	0.7	–0.14	–0.23	0.88	0.84
QN	50%	Y	0.34	0.38	–0.22	–0.21	0.8	0.81
QN	90%	Y	0.64	0.64	–0.09	–0.14	0.96	0.89
Average			0.21	0.28	–0.24	–0.17	0.41	0.54
Standard deviation			0.20	0.24	0.17	0.11	0.33	0.26

assortativity values close to -1 .⁴ The eigenvector centrality monitors the relevance of each node as a function of the relevance of neighbouring nodes. We consider two versions: the first is based on the non-normalized adjacency matrix, the second is based on the row-normalized adjacency matrix. We also normalize the eigenvector centrality of each node to the maximum score obtained by the most connected node. The eigenvector centrality value thus ranges from 0 to 1 and can easily be compared across networks. In the summary tables, we focus on the average eigenvector centrality across nodes. Changes in the eigenvector centrality point out changes in the network structure. Appendix E provides additional details on the considered network measures. Table 1 reports summary network measures for the banks dataset.⁵ Table 1 shows that, on average, the density is slightly smaller during the crisis, whereas the disassortative behaviour (observed in both periods) is greater. Granger causality networks (with or without the common factor) have small densities, that remain almost constant in the two samples, and a disassortative behaviour. Moreover, its average eigenvector centrality increases during the period 2011–2015. The values for the quantile-based networks are quite different, as the previous graphical analysis suggests. In particular, the networks derived from the nonparametric quantile causality test have the greatest density, combined with a disassortative pattern. These two elements could explain the core–periphery structure depicted in Fig. 1.

A large density, combined with a disassortative pattern, is also associated with large eigenvector centrality average values, which indicate that the nodes relevance is quite evenly distributed. This is a somewhat expected result given that we focus on the largest banks. Notably, the picture is quite different in the Granger causality networks, which are characterized by a lower average eigenvector centrality. From a systemic risk perspective, this analysis suggests that the Granger causality provides networks with few relevant nodes. In contrast, the nonparametric quantile causality test provides denser networks, in which the risk is more evenly distributed across nodes, many of which (more than those emerging from Granger causality) are systemically relevant. As the graphs suggest, the nonparametric quantile causality networks also differ from those based on other quantile-based approaches. In particular, the baseline quantile provides summary measures that more closely resemble those of Granger causality. In contrast, the quantile-on-quantile approach produces density measures that are greater (lower) than the ones obtained from the Granger causality (nonparametric quantile causality). We note some interesting differences across quantiles, with larger densities on the extreme quantiles (10% and 90%) than on the median, which comes the closest to the Granger causality. Notably, this would indicate that the network structure changes if we move away from the mean (median). This is an important aspect to bear in mind if our purpose is to analyse the spread of risk during periods of market turmoil. Finally, the introduction of a common market factor seems to have a limited impact.

The results obtained from the insurance companies dataset (see Table F.1 in Appendix F) are similar to the ones we observed for the 25 banks (thus confirming our previous comments), with relevant changes in the network structure when moving from Granger causality to quantile causality. The most relevant difference concerns the network density, which is much larger in the

⁴ The few blank spaces appearing in the following tables of network summary measures correspond to indeterminate forms. The indeterminate forms of the assortativity and eigenvector centrality measures are discussed in Appendix E.

⁵ We report similar tables for both the insurance companies and the 48 Industry Portfolios in Appendix F.

Table 2

The table reports the Jaccard index (scaled from 0 to 100) computed for each pair of networks sorted by row and by column. We denote these networks as follows: QB for baseline quantile regression causality, Qo for quantile-on-quantile causality, QN for nonparametric quantile causality, and GR for Granger causality. The networks generated from the causality in quantiles are estimated at three quantile levels: $\tau = 0.1$ (left panel), $\tau = 0.5$ (central panel) and $\tau = 0.9$ (right panel). We report the results obtained from the Banks (panel (a)), Insurance companies (panel (b)) and Industry portfolios (panel (c)), for the periods 2006–2008 (above the main diagonal of each submatrix) and 2011–2015 (below the main diagonal of each submatrix).

	QB	Qo	QN	GR	QB	Qo	QN	GR	QB	Qo	QN	GR
	$\tau = 0.1$				$\tau = 0.5$				$\tau = 0.9$			
(a) 25 Banks												
QB	–	4.670	2.560	4.760	–	1.560	2.530	11.360	–	11.540	13.220	21.370
Qo	0.430	–	16.330	5.060	1.720	–	5.940	5.830	9.060	–	18.130	1.750
QN	4.200	28.200	–	11.940	5.680	2.400	–	3.780	11.300	22.200	–	11.990
GR	12.960	5.810	6.940	–	16.670	0.000	7.690	–	10.090	5.060	7.930	–
(b) 25 Insurance companies												
QB	–	3.600	7.810	4.610	–	1.820	5.970	8.970	–	2.740	3.930	3.730
Qo	2.780	–	15.980	12.180	0.910	–	1.560	3.880	10.600	–	21.920	5.110
QN	4.190	19.610	–	14.570	11.700	4.500	–	10.030	5.390	15.940	–	14.600
GR	6.000	4.350	0.850	–	6.250	2.270	0.960	–	16.330	5.800	1.650	–
(c) 48 Industry portfolios												
QB	–	11.260	6.840	10.090	–	8.160	13.690	34.460	–	7.260	14.050	25.250
Qo	4.660	–	15.910	7.470	4.280	–	7.700	8.190	8.960	–	12.430	6.870
QN	4.830	32.220	–	11.440	4.630	5.920	–	10.280	7.710	33.170	–	12.900
GR	6.540	1.260	2.590	–	3.700	2.440	3.400	–	8.290	1.530	2.200	–

tails than in the mean (which coincides with Granger causality). Moreover, the nonparametric quantile causality provides the greatest average eigenvector centrality. Likewise, when using the 48 Industry Portfolios dataset (see Table F.2 in Appendix F), we find relevant differences between the Granger and the quantile causality networks. This holds particularly for network density and average eigenvector centrality. However, we note that the structure of the nonparametric quantile causality networks differs considerably from the other quantile causality networks for all three datasets. To summarize, the findings discussed above confirm that the different methods used to quantify the causality relations (i.e., GR, QB, Qo and QN) lead to important changes in the network structure.

Our method points at combining different networks, each of which should reflect a different view on causality. Therefore, before moving to the network combination, we analyse the similarity of the individual networks. Indeed, combining very similar networks might be of little interest, while we are most interested in combining networks carrying different information. We measure the similarity of different networks with the Jaccard index (Jaccard, 1901), which takes values between 0 (maximum discordance) and 100 (perfect matching). We report the results in Table 2.⁶ In general, the Jaccard index takes low values for all the network pairs, thus highlighting the usefulness of network combination. We highlight that the networks based on nonparametric quantile regression and those derived from quantile-on-quantile tend to be more similar; see, e.g., the case of the 48 Industry Portfolios, for $\tau = 0.1$ and for the period 2011–2015, where the Jaccard index is equal to 32.22. In contrast, the baseline quantile regression networks tend to be closer to the Granger causality networks; see, e.g., the case of the 48 Industry Portfolios, for $\tau = 0.5$ and the period 2006–2008, where the Jaccard index is equal to 34.46. Given these evidences, supporting the different views provided by alternative causality networks, we proceed with the estimation of the composite network.

3.3. The composite causality network

Starting from the availability of causality networks derived from four different methods – QB, Qo, QN and GR (see Section 3.2) – and over three quantile levels (10%, 50% and 90%), we now estimate the composite networks. We estimate the model in (4), accounting for the presence of common factors, for which we follow the usual practice and use the market factor, the size factor, the book-to-market factor and the momentum factor, following Fama and French (1993), Fama and French (1995) and Carhart (1997). A priori, we have no information to suggest a possible preference for a particular network. We then estimate the composite network starting from four different layers, where the quantile-based networks are associated with the same reference quantile. We provide a sensitivity analysis on the effects of excluding a single network (among the four available layers) in Section 3.4, where we report a set of robustness checks. When estimating factor models, it is common to use a monthly frequency. However, this would leave us with only 36 (60) observations in the first (second) sample. We then estimate the linear factor model (augmented with network dependencies) by using weekly returns. This choice allows us to increase the number of observations—108 in the first sample and 316 in the second one. Moreover, this choice has advantages in terms of model estimation and parameter inference. We do not consider daily returns as they would require the introduction of heteroskedastic dynamics in the variance of the residuals, increasing the complexity of the estimation due to the well-known curse of dimensionality. Estimates of the network combination on monthly returns are nonetheless provided in Section 3.4 and in Appendix G.

⁶ We report in Table 2 the results obtained by estimating the four causality networks without a common factor. The results we obtain when including the common factor, that we do not report here for the sake of space, are similar and available upon request.

Table 3

The table reports the estimates of the parameters δ_i , that represent the weights for networks combination. We report the estimates obtained by using the banks (top panel), the insurance companies (middle panel) and the 48 industry portfolios (bottom panel) datasets. The first column reports the quantile levels used to estimate the quantile-based networks. The second column indicates whether a common factor has been used (Y) or not (N) in the estimation of the causality networks. Columns 3–6 (7–10) refer to the period 2006–2008 (2011–2015). The second row identifies the four competing networks, that we optimally combine, derived from the following methods: baseline quantile regression (QB), quantile-on-quantile causality (Qo), nonparametric quantile causality (QN) and Granger causality (GR). The reported parameters are, by construction, positive and sum up to one (within each row and within each period). A star points out a statistical significance of the estimates at the 5% level.

Quantile	Factor	2006–2008				2011–2015			
		QB	Qo	QN	GR	QB	Qo	QN	GR
		25 Banks							
10%	N	0.061	0.055	0.875	0.010	0.000	0.630*	0.354*	0.017*
50%	N	0.146	0.108	0.736*	0.010	0.091*	0.065*	0.773*	0.071
90%	N	0.005	0.008	0.883	0.104	0.000	0.706*	0.243	0.051*
10%	Y	0.022	0.073	0.827*	0.077	0.000	0.538*	0.428*	0.034*
50%	Y	0.052	0.057	0.842*	0.050	0.107	0.072*	0.649*	0.173*
90%	Y	0.134	0.000*	0.762*	0.104*	0.000	0.876*	0.121	0.003*
		25 Insurance companies							
10%	N	0.000	0.022*	0.951*	0.027	0.058	0.831	0.111	0.000
50%	N	0.865*	0.000	0.032*	0.103	0.059	0.045	0.886*	0.010
90%	N	0.000	0.028	0.972*	0.000	0.000	0.364*	0.436*	0.201
10%	Y	0.001	0.006	0.948*	0.045	0.000	0.488	0.512	0.000
50%	Y	0.731*	0.054	0.188*	0.027*	0.051	0.040	0.837*	0.072
90%	Y	0.000	0.046	0.954*	0.000	0.000	0.554	0.446	0.000
		48 Industry portfolios							
10%	N	0.000	1.000*	0.000	0.000	0.032	0.968*	0.000	0.000
50%	N	0.212	0.000	0.478	0.310	0.021	0.031*	0.848*	0.100
90%	N	0.108	0.892*	0.000	0.000	0.000	1.000*	0.000	0.000
10%	Y	0.178	0.471	0.000	0.351	0.036	0.695*	0.228	0.041
50%	Y	0.123	0.000	0.000	0.877*	0.092	0.007*	0.836*	0.064
90%	Y	0.000	0.810	0.000	0.190	0.107	0.808	0.030	0.055

The estimates of the linear factor model with network dependencies generate different outputs. First, we analyse the model parameters. We refer to both the weight of each network, as measured by the δ_i parameters, and the impact of the composite network on the assets returns, i.e., the parameters included in $\mathcal{R}$. Second, we focus on the composite network, obtained by combining the primitive networks weighted by the δ_i coefficients. Third, we study the model residuals, which contain useful information for assessing the advantages of moving to a composite network, as opposed to the benchmark cases—absence of network dependencies (where the contemporaneous link matrix A is an identity matrix) or network dependencies captured by Granger causality. The latter is a valuable benchmark as we would like to highlight the potential improvements of measuring the links among assets in their quantiles, going beyond the mean. We compare the estimated models taking into account the average correlation of their residuals. We obtain improvements when the residuals have a smaller correlation level, that is, more whitened residuals. Table 3 reports the parameters estimated for the combined network using the three datasets and over the periods 2006–2008 and 2011–2015. Bearing in mind that the sum of these parameters is equal to 1 and that they take positive values, a larger coefficient points out a greater relevance of the associated network. We also assess the significance of the estimated parameters. Focusing on the banks dataset, we note that the nonparametric quantile causality network is the most relevant during the financial crisis period and across all quantiles. Notably, the estimated δ parameter for this network ranges between 0.736 and 0.883. Furthermore, it is always statistically significant when the estimation of the networks is made with the presence of a common market factor.

We stress that, from a systemic risk perspective, this result suggests that the risk is more widespread across banks, as the nonparametric quantile network is much denser than the other causality networks. This greater relevance of the nonparametric quantile causality network also emerges for the median and for the lower and upper tails. In all cases, the Granger causality network has a very small weight (which is non-significant in five cases out of six). When focusing on the period 2011–2015, both the nonparametric quantile and the quantile-on-quantile causality networks are relevant, with and without the presence of common market factors. Here again, although it is more statistically significant, the Granger causality network has smaller weights than the quantile-based causality networks. Overall, in both samples, the baseline quantile regression network has the least relevant role. Finally, in the second sample, we see a clear change in the combined network parameters when moving from median networks to networks estimated on either the lower or upper tails. In particular, the impact of the quantile-on-quantile networks increases, thus supporting the importance of looking at quantile causality too when accounting for interdependence across assets in a linear factor model and showing that a double conditioning (on both the response and the explanatory variables) is important when estimating causality at quantile levels.

The insurance companies dataset produces similar results. In both periods, Granger causality networks are the least relevant, being associated with small coefficients (and only one out of twelve is statistically significant). The nonparametric quantile causality is the most relevant during the period 2006–2008, although the baseline quantile causality network has a larger weight in the median case (unlike the picture emerging from the banks dataset). Both the nonparametric quantile causality and the quantile-on-quantile

Table 4

The table reports summary measures of the composite networks estimated from (4). We build the composite networks by combining four methods: the baseline quantile regression, the quantile-on-quantile, the nonparametric quantile and the Granger causalities. We report the results obtained by using the banks (top panel), the insurance companies (middle panel) and the 48 industry portfolios (bottom panel) datasets. The first column reports the quantile levels used to estimate the quantile-based networks. The second column indicates whether a common factor has been used (Y) or not (N) in the estimation of the causality networks. We report the results obtained for the periods 2006–2008 and 2011–2015.

Measures		Density		Assortativity		Eigenvector centrality		Eigenvector centrality	
								Adj weighted	
Quantile	Factor	06–08	11–15	06–08	11–15	06–08	11–15	06–08	11–15
25 Banks									
10%	N	0.63	0.79	–0.30	–0.23	0.70	0.83	0.35	0.47
50%	N	0.45	0.43	–0.28	–0.28	0.62	0.61	0.24	0.43
90%	N	0.69	0.76	–0.32	–0.25	0.70	0.77	0.32	0.57
10%	Y	0.65	0.87	–0.36	–0.25	0.67	0.87	0.39	0.64
50%	Y	0.45	0.48	–0.31	–0.36	0.65	0.69	0.31	0.41
90%	Y	0.72	0.85	–0.28	–0.25	0.75	0.85	0.39	0.64
25 Insurance companies									
10%	N	0.88	0.72	–0.17	–0.29	0.88	0.74	0.66	0.45
50%	N	0.59	0.61	–0.22	–0.21	0.72	0.85	0.37	0.60
90%	N	0.87	0.72	–0.15	–0.24	0.91	0.77	0.66	0.50
10%	Y	0.82	0.82	–0.22	–0.30	0.85	0.82	0.55	0.59
50%	Y	0.66	0.64	–0.25	–0.23	0.79	0.84	0.55	0.51
90%	Y	0.83	0.83	–0.22	–0.26	0.83	0.83	0.60	0.59
48 Industry portfolios									
10%	N	0.71	0.74	–0.22	–0.26	0.77	0.78	0.46	0.62
50%	N	0.50	0.37	–0.30	–0.29	0.58	0.51	0.29	0.19
90%	N	0.71	0.74	–0.21	–0.24	0.76	0.76	0.38	0.62
10%	Y	0.83	0.89	–0.24	–0.24	0.83	0.90	0.51	0.69
50%	Y	0.53	0.42	–0.35	–0.37	0.56	0.55	0.24	0.23
90%	Y	0.80	0.92	–0.29	–0.15	0.80	0.92	0.47	0.70

causality are relevant in the second sample (2011–2015), with changes in the estimated parameters when moving from networks estimated on the median to networks estimated on either the 10% or 90% quantiles. Finally, the results are more heterogeneous when using the 48 industry portfolios dataset in the first sample (2006–2008); in contrast, they are consistent with the two previous cases in the period 2011–2015. In the first period, the significance is limited in many cases, mainly for the quantile-based causality networks estimated at the median. In the second period, there is a marked difference between the quantile-based causality estimated at the median and those estimated on the tails. Be that as it may, the Granger causality networks have the smallest weight (and are never statistically significant).

Table 4 reports the summary measures for the combined networks. The various composite networks are similar in all the quantities we report (for a given sample period and a given dataset). The identified heterogeneity is much smaller than the one observed for the primitive networks. Figs. 3–5 display the combined networks for the period 2006–2008.⁷ We link this finding to the use of a linear factor model augmented with network dependencies. The contemporaneous relations among the modelled variables capture their correlations and go beyond what we associate with common factors. It might be that the various composite networks capture the dependencies among the returns (beyond common factors) in a similar way. The differences we find depend partly on the heterogeneity across networks and partly on the different weights assigned to the primitive networks.

To shed further light on the differences between the composite networks as well as on the improvements gained by the linear factor model augmented with network combination, we provide in Tables 5–7 descriptive analyses of the correlations among the residuals computed from (4). Tables 5–7 include two benchmarks: (i) a standard linear factor model, that is, the (Carhart, 1997) 4-factors CAPM; and (ii) a linear factor model augmented with network dependencies, where the network is derived from the Granger causality (with or without the inclusion of a common factor in estimating causal relationships). The results obtained by using the banks and the insurance companies datasets are similar. The residuals of the 4-factors CAPM have the largest median correlations, with a distribution of these correlations shifted to the right and a clear prevalence of positive values. The introduction of network dependencies, when measured by Granger causality, improves the model fit, but the predominance of positive values in the correlation remains (see the small fraction of correlations below –0.1), and the median correlation remains above 0.1. The model fit improves considerably with the introduction of a plurality of networks. Indeed, the residual correlations are centred at zero, with a larger fraction of correlations below –0.1 and a marked reduction of large positive residual correlations. As a result, we conclude that our approach based on combining several networks within a network augmented linear factor model overperforms the standard Granger causality. The improvements achieved by the latter, over the multifactor model, were already documented in Billio et al. (2017).

⁷ All the other combined network visualizations are in Appendix H.

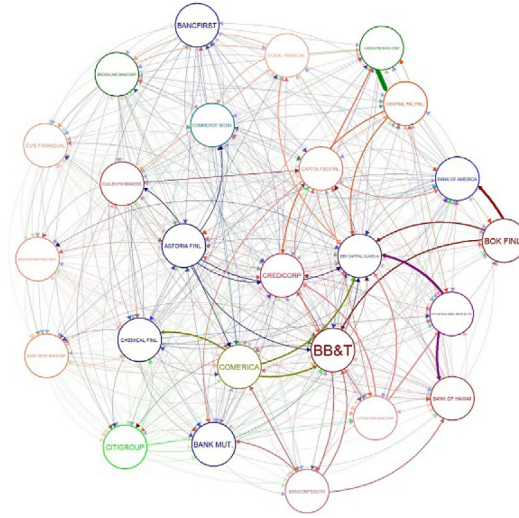


Fig. 3. The figure displays the network for the 25 banks dataset. The network is obtained by combining the baseline quantile regression (QB), the quantile-on-quantile (Qo), the nonparametric quantile (QN) – at the 10% quantile – and the Granger causality methods, during the period 2006–2008. Here, we do not use common factors for network estimations.

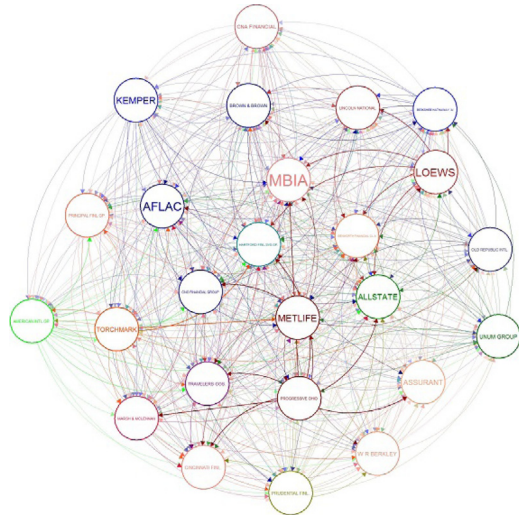


Fig. 4. The figure displays the network for the 25 insurance companies dataset. The network is obtained by combining the baseline quantile regression (QB), the quantile-on-quantile (Qo), the nonparametric quantile (QN) – at the 10% quantile – and the Granger causality methods, during the period 2006–2008. Here, we do not use common factors for network estimations.

The results are less clear when using the 48 industry portfolios dataset, as the competing approaches provide similar findings. This is in line with the heterogeneous results seen for the weights of the various networks, where we were unable to identify a clear preference. We interpret this as an evidence to suggest that network augmented linear factor models might improve traditional linear factor models when using single assets and not industry portfolios. The aggregation of assets into portfolios probably distorts the dependence structure across assets and limits its impact. There might be a different reason linking the datasets of the banks and insurance companies to the potential presence of a sector-specific factor in the analysis. Nevertheless, in this case, we should have seen no improvement in the residual correlations of the composite network. We then believe that the idea of a missing factor is not consistent with our findings. Further analyses are needed on this topic, but we leave them to future research.

We close this section with a few comments on the coefficients for monitoring the impact of the composite networks on the various assets, as included in the diagonal of matrix R in (4). Appendix I reports the plots of the coefficients for each bank, insurance company and industrial portfolio, across the various combined networks. Overall, we note that the coefficients are usually positive and significant for banks and insurance companies, whereas the network impact is more heterogeneous and of limited significance for industrial portfolios. These results deserve a more thorough analysis, but this goes beyond the scope of the present paper.

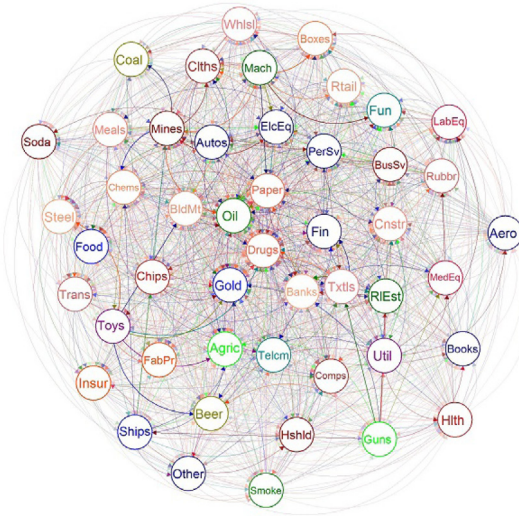


Fig. 5. The figure displays the network for the 48 industry portfolios dataset. The network is obtained by combining the baseline quantile regression (QB), the quantile-on-quantile (Qo), the nonparametric quantile (QN) – at the 10% quantile – and the Granger causality methods, during the period 2006–2008. Here, we do not use common factors for network estimations.

Table 5

The table reports residual correlation descriptive analyses for the 25 banks dataset. The first column identifies the various models, whereas the second column indicates the number of networks used in the model. In the first column, $Q(10\%)$ denotes the use of the combination of networks derived from the following methods: baseline quantile regression (QB), quantile-on-quantile (Qo), nonparametric quantile (QN) and Granger (GR) causalities; QB, Qo and QN are estimated at the quantile level of 0.1. Similar considerations hold for the quantile levels of 0.5 and 0.9. GR denotes the model which uses only the Granger causality network, whereas the last row refers to the 4-factor CAPM (Multifactor). The table reports the following statistics of the residuals correlations: the minimum (Min), the maximum (Max), the 0.1-th quantile (q10), the median (q50), the 0.9-th quantile (q90) and the number of elements in the correlation matrix lower than or equal to -0.1 . The third column indicates whether a common factor has been used (Y) or not (N) in the estimation of the model.

Model	Number of networks	Factor	Min	Max	q10	q50	q90	% elements ≤ -0.1
Period 2006–2008								
Q(10%)	4	N	−0.279	0.584	−0.106	0.071	0.282	11.7%
Q(50%)	4	N	−0.345	0.596	−0.109	0.062	0.311	10.7%
Q(90%)	4	N	−0.269	0.600	−0.146	0.050	0.276	16.0%
Q(10%)	4	Y	−0.297	0.566	−0.135	0.035	0.238	17.7%
Q(50%)	4	Y	−0.349	0.503	−0.119	0.047	0.242	14.7%
Q(90%)	4	Y	−0.297	0.455	−0.161	0.020	0.218	18.3%
GR	1	N	−0.381	0.670	−0.128	0.111	0.372	13.3%
GR	1	Y	−0.394	0.641	−0.128	0.089	0.378	11.7%
Multifactor	-	-	−0.358	0.678	−0.108	0.257	0.502	10.7%
Period 2011–2015								
Q(10%)	4	N	−0.263	0.422	−0.122	−0.013	0.135	15.3%
Q(50%)	4	N	−0.397	0.537	−0.129	0.055	0.224	13.7%
Q(90%)	4	N	−0.253	0.431	−0.127	0.010	0.138	15.7%
Q(10%)	4	Y	−0.283	0.503	−0.132	−0.008	0.140	19.0%
Q(50%)	4	Y	−0.330	0.505	−0.126	0.041	0.197	14.7%
Q(90%)	4	Y	−0.304	0.508	−0.140	0.001	0.132	17.7%
GR	1	N	−0.211	0.598	−0.027	0.151	0.362	3.0%
GR	1	Y	−0.211	0.598	−0.027	0.146	0.361	3.3%
Multifactor	-	-	−0.122	0.669	0.054	0.266	0.487	0.3%

We further evaluate our combining method with additional analyses under both a statistical and a financial viewpoint. First, we estimate a four-factor model, as defined in (4), for each company or industry portfolio in our dataset. We use the standard common factors: (i) the market excess return (MKT); (ii) the “small minus big” market capitalization (SMB); (iii) the “high minus low” book-to-market ratio (HML); and (iv) the “momentum” (MOM). We estimate the model in (4) by combining four different networks W_j , building on both the causality in quantiles and the standard Granger causality. We then check whether the relations in quantiles provide additional information with respect to the standard causality in mean. Moreover, we use a leave-one-out technique, that is, excluding, from time to time, an individual network from the combination. This allows us to evaluate the loss we incur when we exclude a particular method designed to measure causality in quantiles. In particular, we use the following combinations: (i) $Q(\tau)$, that combines the baseline quantile (QB), the quantile-on-quantile (Qo), the nonparametric quantile (QN) and the Granger (GR)

Table 6

The table reports residual correlation descriptive analyses for the 25 insurance companies dataset. The first column identifies the various models, whereas the second column indicates the number of networks used in the model. In the first column, $Q(10\%)$ denotes the use of the combination of networks derived from the following methods: baseline quantile regression (QB), quantile-on-quantile (Qo), nonparametric quantile (QN) and Granger (GR) causalities; QB, Qo and QN are estimated at the quantile level of 0.1. Similar considerations hold for the quantile levels of 0.5 and 0.9. GR denotes the model which uses only the Granger causality network, whereas the last row refers to the 4-factor CAPM (Multifactor). The table reports the following statistics of the residuals correlations: the minimum (Min), the maximum (Max), the 0.1-th quantile (q10), the median (q50), the 0.9-th quantile (q90) and the number of elements in the correlation matrix lower than or equal to -0.1 . The third column indicates whether a common factor has been used (Y) or not (N) in the estimation of the model.

Model	Number of Networks	Factor	Min	Max	q10	q50	q90	% elements ≤ -0.1
Period 2006–2008								
Q(10%)	4	N	−0.437	0.686	−0.205	0.003	0.250	27.7%
Q(50%)	4	N	−0.429	0.669	−0.167	0.039	0.258	15.3%
Q(90%)	4	N	−0.448	0.676	−0.207	0.009	0.255	25.7%
Q(10%)	4	Y	−0.436	0.684	−0.203	0.004	0.246	28.0%
Q(50%)	4	Y	−0.417	0.473	−0.176	0.023	0.226	19.3%
Q(90%)	4	Y	−0.464	0.671	−0.214	0.001	0.253	27.0%
GR	1	N	−0.466	0.831	−0.166	0.044	0.316	17.7%
GR	1	Y	−0.460	0.714	−0.166	0.048	0.323	15.0%
Multifactor	-	-	−0.370	0.847	−0.149	0.089	0.422	14.0%
Period 2011–2015								
Q(10%)	4	N	−0.290	0.504	−0.121	0.009	0.157	14.7%
Q(50%)	4	N	−0.258	0.548	−0.149	0.019	0.202	16.7%
Q(90%)	4	N	−0.289	0.348	−0.137	−0.006	0.167	19.3%
Q(10%)	4	Y	−0.333	0.503	−0.153	−0.012	0.189	20.7%
Q(50%)	4	Y	−0.267	0.592	−0.156	0.016	0.192	18.3%
Q(90%)	4	Y	−0.299	0.398	−0.152	−0.001	0.199	21.3%
GR	1	N	−0.136	0.658	−0.005	0.121	0.317	1.7%
GR	1	Y	−0.136	0.658	−0.014	0.116	0.315	1.7%
Multifactor	-	-	−0.103	0.658	0.017	0.138	0.332	0.3%

Table 7

The table reports residual correlation descriptive analyses for the 48 industry portfolios dataset. The first column identifies the various models, whereas the second column indicates the number of networks used in the model. In the first column, $Q(10\%)$ denotes the use of the combination of networks derived from the following methods: baseline quantile regression (QB), quantile-on-quantile (Qo), nonparametric quantile (QN) and Granger (GR) causalities; QB, Qo and QN are estimated at the quantile level of 0.1. Similar considerations hold for the quantile levels of 0.5 and 0.9. GR denotes the model which uses only the Granger causality network, whereas the last row refers to the 4-factor CAPM (Multifactor). The table reports the following statistics of the residuals correlations: the minimum (Min), the maximum (Max), the 0.1th quantile (q10), the median (q50), the 0.9th quantile (q90) and the number of elements in the correlation matrix lower than or equal to -0.1 . The third column indicates whether a common factor has been used (Y) or not (N) in the estimation of the model.

Model	Number of networks	Factor	Min	Max	q10	q50	q90	% elements ≤ -0.1
Period 2006–2008								
Q(10%)	4	N	−0.465	0.611	−0.191	−0.001	0.201	25.6%
Q(50%)	4	N	−0.369	0.663	−0.161	0.018	0.249	20.6%
Q(90%)	4	N	−0.443	0.608	−0.195	−0.006	0.206	25.9%
Q(10%)	4	Y	−0.417	0.625	−0.187	0.006	0.220	24.1%
Q(50%)	4	Y	−0.454	0.627	−0.159	0.011	0.233	20.4%
Q(90%)	4	Y	−0.508	0.592	−0.187	−0.007	0.215	23.8%
GR	1	N	−0.384	0.687	−0.146	0.025	0.257	18.6%
GR	1	Y	−0.401	0.687	−0.157	0.011	0.239	19.8%
Multifactor	-	-	−0.486	0.733	−0.219	0.003	0.267	27.1%
Period 2011–2015								
Q(10%)	4	N	−0.463	0.567	−0.140	−0.008	0.135	19.7%
Q(50%)	4	N	−0.454	0.518	−0.136	0.005	0.169	16.7%
Q(90%)	4	N	−0.471	0.572	−0.144	−0.013	0.139	19.9%
Q(10%)	4	Y	−0.432	0.497	−0.147	−0.004	0.148	18.9%
Q(50%)	4	Y	−0.470	0.511	−0.135	0.001	0.155	16.6%
Q(90%)	4	Y	−0.438	0.565	−0.143	−0.006	0.154	17.9%
GR	1	N	−0.470	0.577	−0.140	0.007	0.199	18.0%
GR	1	Y	−0.503	0.577	−0.139	0.006	0.177	17.7%
Multifactor	-	-	−0.456	0.592	−0.147	0.007	0.208	18.4%

causality methods; (ii) $Q(\tau)\backslash QN$, that combines QB, Qo and GR; (iii) $Q(\tau)\backslash Qo$, that combines QB, QN and GR; and (iv) $Q(\tau)\backslash QB$, that combines Qo, QN and GR. Table 8 reports the Bayesian Information criterion (BIC) and the cross-sectional means of the loadings to the common factors for the case of $\tau = 0.1$. Here, we also report the results obtained by using in (4) the network building on

Table 8

The table reports the results obtained by estimating the four-factor model in (4) for each company or industry portfolio in our dataset, by using MKT, SMB, HML and MOM as common factors. We estimate the model in (4) by combining four different networks W_j : (i) Q(10%), that combines the baseline quantile (QB), the quantile-on-quantile (Qo), the nonparametric quantile causality (QN)—QB, Qo and QN are estimated for a quantile level of 0.1—and the Granger causality (GR); (ii) Q(10%)\QN, that combines QB, Qo and GR; (iii) Q(10%)\Qo, that combines QB, QN and GR; (iv) Q(10%)\QB, that combines Qo, QN and GR. We also report the results obtained by using in (4) the network building on the Granger causality only (GR) and by estimating the standard Multifactor model (Multifactor). The second column indicates whether a common factor has been used (Y) or not (N) in the estimation of the causality networks. From left to right we report the Bayesian information criterion (BIC) and the cross-sectional means of the coefficients multiplying, respectively, MKT, SMB, HML and MOM. We report the results obtained for the banks (panel (a)), insurance companies (panel (b)) and industry portfolios (panel (c)) dataset, for both the periods 2006–2008 and 2011–2015.

Model	Factor	Period 2006–2008					Period 2011–2015				
		BIC	MKT	SMB	HML	MOM	BIC	MKT	SMB	HML	MOM
(a) 25 Banks											
Q(10%)	N	14,623.02	0.487	0.501	0.603	−0.214	15,597.39	0.328	0.131	0.290	−0.014
Q(10%)\QN	N	14,873.10	0.567	0.533	0.739	−0.265	15,616.02	0.328	0.133	0.293	−0.017
Q(10%)\Qo	N	14,720.47	0.473	0.462	0.643	−0.227	15,883.49	0.441	0.169	0.393	0.002
Q(10%)\QB	N	14,630.00	0.504	0.482	0.630	−0.218	15,586.27	0.328	0.131	0.290	−0.014
Q(10%)	Y	14,500.02	0.417	0.401	0.553	−0.182	15,621.93	0.321	0.134	0.296	−0.009
Q(10%)\QN	Y	14,682.45	0.455	0.456	0.539	−0.160	15,641.18	0.302	0.133	0.283	−0.016
Q(10%)\Qo	Y	14,576.26	0.417	0.365	0.612	−0.212	15,835.50	0.421	0.163	0.378	0.006
Q(10%)\QB	Y	14,496.45	0.432	0.404	0.566	−0.185	15,610.81	0.321	0.134	0.296	−0.009
GR	N	15,036.01	0.621	0.543	0.842	−0.290	16,825.94	0.826	0.309	0.742	0.007
GR	Y	14,992.05	0.591	0.464	0.835	−0.278	16,820.78	0.818	0.306	0.737	0.008
Multifactor	–	–	0.792	0.634	1.012	−0.379	–	1.041	0.386	0.927	0.021
(b) 25 Insurance companies											
Q(10%)	N	15,951.60	0.561	−0.275	0.515	−0.135	14,822.99	0.447	0.046	0.283	−0.022
Q(10%)\QN	N	16,098.24	0.799	−0.385	0.596	−0.163	14,818.14	0.461	0.047	0.290	−0.025
Q(10%)\Qo	N	15,953.61	0.557	−0.282	0.503	−0.133	15,183.67	0.656	0.054	0.400	0.010
Q(10%)\QB	N	15,962.14	0.575	−0.272	0.533	−0.144	14,817.68	0.448	0.046	0.285	−0.018
Q(10%)	Y	15,974.15	0.575	−0.279	0.511	−0.138	14,936.75	0.433	0.039	0.260	−0.009
Q(10%)\QN	Y	16,126.77	0.924	−0.416	0.681	−0.180	14,948.24	0.414	0.040	0.245	−0.016
Q(10%)\Qo	Y	15,960.33	0.573	−0.280	0.508	−0.137	15,144.31	0.626	0.048	0.382	0.009
Q(10%)\QB	Y	15,964.07	0.575	−0.279	0.511	−0.138	14,925.62	0.433	0.039	0.260	−0.009
GR	N	16,230.68	0.957	−0.471	0.747	−0.205	15,647.27	1.009	0.085	0.607	0.017
GR	Y	16,214.02	0.989	−0.475	0.744	−0.201	15,643.71	1.000	0.084	0.600	0.017
Multifactor	–	–	1.195	−0.561	1.024	−0.319	–	1.061	0.066	0.624	0.037
(c) 48 Industry portfolios											
Q(10%)	N	18,082.31	0.680	0.163	0.039	−0.010	21,726.09	0.562	0.109	−0.005	−0.063
Q(10%)\QN	N	18,070.58	0.685	0.165	0.039	−0.012	21,714.96	0.562	0.109	−0.005	−0.063
Q(10%)\Qo	N	18,275.40	1.062	0.248	0.054	0.039	22,150.26	0.761	0.142	0.005	−0.071
Q(10%)\QB	N	18,056.33	1.030	0.233	0.047	0.021	21,728.45	0.554	0.107	−0.005	−0.063
Q(10%)	Y	18,111.02	0.664	0.149	0.067	0.029	21,954.41	0.559	0.122	0.010	−0.066
Q(10%)\QN	Y	18,100.92	0.664	0.149	0.067	0.029	21,972.34	0.556	0.125	0.009	−0.068
Q(10%)\Qo	Y	18,157.25	0.972	0.217	0.069	0.030	22,126.76	0.669	0.131	−0.003	−0.066
Q(10%)\QB	Y	18,121.19	0.783	0.180	0.060	0.024	21,949.03	0.572	0.124	0.011	−0.066
GR	N	18,343.75	1.288	0.303	0.063	0.042	22,612.23	0.983	0.180	−0.005	−0.094
GR	Y	18,287.65	1.180	0.264	0.066	0.039	22,506.87	0.952	0.174	−0.011	−0.092
Multifactor	–	–	1.081	0.257	0.029	0.000	–	1.023	0.171	−0.003	−0.081

the Granger causality only (GR) and by estimating the standard multifactor model (Multifactor). The results for $\tau = \{0.5, 0.9\}$ are reported in Appendix F.

When analysing the BIC, we note that our combinations outperform GR. This highlights the advantages of using the causality in quantiles rather than the Granger causality when building financial networks. The most general combination, Q(τ), is always placed among the top-ranked approaches, highlighting the synergies coming from QB, Qo, QN and GR when combined in a unique structure. When using the leave-one-out technique to compare the individual performance of QB, Qo and QN, respectively, we check that QN provides the greatest contribution for banks and insurances during the period 2006–2008, characterized by a higher risk during the subprime crisis. In contrast, Qo provides a better performance during the period 2011–2015 for $\tau = \{0.1, 0.9\}$ (see Table 8 and Appendix F (Table F.4)), whereas QN becomes relevant for $\tau = 0.5$ (see Appendix F (Table F.3)).

We now analyse the cross-sectional means of the factor loadings; see Table 8 for $\tau = 0.1$.⁸ In general, banks insurances and industry portfolios have a greater sensitivity to the four common factors when using GR and Multifactor. This is expected, as the knowledge of the networks, thanks to the model defined in (4), permits to filter out the indirect exposure to common factors endogenously transmitted by network links, and captures the structural asset exposure to the common factors. The latter is usually lower than the exposure we observe by neglecting the network information when the model coefficients R in (4) are positive (see

⁸ The results for $\tau = \{0.5, 0.9\}$ are reported in Appendix F.

Appendix (i). Indeed, the most general combination $Q(\tau)$ provides, on average, the lowest sensitivity to MKT, SMB and HML, and this holds for all quantile levels τ (see Table 8 and Appendix F). For the MOM factor, we note that in the second period, and for Industry also in the first period, the loadings are very close to zero.

The leave-one-out exercise highlights that the exclusion of QN leads to an increase in the sensitivity to MKT for banks and insurances. This result is more evident during the period 2006–2008. Therefore, the individual companies seem more exposed to the market risk when excluding QN, but this is only apparent as part of the exposure is mediated by the assets interconnections. The trend is less clear during the period 2011–2015. Here, we observe cases in which other quantile-based networks seem to provide a greater impact on factor exposures. The impact of the combined networks changes over asset classes as well as over periods and quantiles (see Table 8 and Appendix F).

We also evaluate the contribution of our network combination approach from a financial viewpoint, that is, in terms of portfolio allocation. In particular, we analyse the performance of two different strategies. The first, SA, is made up with long positions on stocks with lower centrality and short positions on stocks with higher centrality. The second, SB, is the opposite of SA: it is long on stocks with high centrality and short on stocks with low centrality. We use these strategies to investigate when the centrality improves the portfolio performance, according to crisis, boom and calm periods. Higher centrality stocks could be difficult to diversify, because they are connected to other stocks, inflating the idiosyncratic portfolio variance. In contrast, lower centrality stocks could be useful, as they are weakly correlated to other stocks through the links in the estimated network. For each of our datasets, we sort the stocks according to their centrality and cluster them in quintiles. We then implement SA and SB by using only the stocks included in the first and in the fifth quintiles. In particular, SA is made up with equally weighted long (short) positions on the stocks included in the first (fifth) quintile. The opposite holds for SB. For both SA and SB, the number of stocks in either the first or the fifth quintile depends on the dataset we are using: five for banks and insurances and ten for the 48 industry portfolios. For both SA and SB, the stocks centrality is retrieved from the networks we defined above: $Q(\tau)$, $Q(\tau) \setminus QN$, $Q(\tau) \setminus Qo$, $Q(\tau) \setminus QB$ and GR. We then compare the contribution of each network to the performance of the resulting portfolio. Furthermore, we compare SA and SB with the equally weighted portfolio (EW); see, e.g., DeMiguel et al. (2009), Duchin and Levy (2009) and Tu and Zhou (2011). We compare the competing strategies by using three performance indicators: the standard deviation of the portfolio returns (STD), the Sharpe ratio (SR) and the Jensen's α retrieved from the Fama and French (1993)'s model estimated for each portfolio strategy ($\hat{\alpha}$). Moreover, we also check whether the variances and the Sharpe ratios of both SA and SB are statistically different from the ones generated by EW by using the tests proposed by Ledoit and Wolf (2008, 2011). We report in Table 9 the results obtained at $\tau = 0.1$. For the sake of space, we report the results obtained at $\tau = \{0.5, 0.9\}$ in Tables J.1–J.2 (see Appendix J). However, we discuss below the findings retrieved from a joint analysis of all three quantile levels— $\tau = \{0.1, 0.5, 0.9\}$.

Table 9 (as well as the additional results in Appendix J) points out that the most general combination, $Q(\tau)$, is always placed among the top-ranked strategies for all the performance indexes. Again, this highlights the synergies coming from QB, Qo, QN and GR, when combined in a unique network. SA and SB provide the same performance in terms of STD, as one of them is the opposite of the other and the differences disappear when computing squared returns. When evaluating the individual contribution of the three causality networks (QB, Qo and QN) by using the leave-one-out procedure, we check that none of the three methodologies systematically dominates the others. In contrast, the results change according to the performance index (STD, SR or $\hat{\alpha}$), the dataset (banks, insurances or industry portfolios), the strategy (SA or SB) and the period (2006–2008 or 2011–2015).

Finally, we compare the strategies building on the causality in quantiles networks with the one building on the Granger causality network (GR). Many of the strategies retrieved from the causality in quantiles provide a lower standard deviation in the period 2006–2008, where the risk minimization acquires a critical importance due to the market turmoil generated by the subprime crisis. This might be linked to the greater flexibility of these methods compared to the Granger causality. In contrast, GR provides a lower standard deviation during the period 2011–2015, that is, during a recovery phase in which it is possible to implement more aggressive strategies, also thanks to the sparsity of the Granger network. We read the latter as a consequence of the reduced perception of systemic risks during the recovery phase.⁹ When using the banks dataset for $\tau = \{0.1, 0.9\}$, GR tends to have a better performance in terms of Sharpe ratio in period 2006–2008. In contrast, many of the networks building on the causality in quantiles have a better performance when using the insurances dataset in the same period. Moving to the period 2011–2015, many of the networks obtained from the causality in quantiles provide a better performance in terms of Sharpe ratio. Similar results hold when analysing $\hat{\alpha}$. When we compare the strategies building on the causality in quantiles with EW, Table 9 as well as the additional results in Appendix J show that the strategies building on our combination (for both SA and SB) outperform EW in most of the cases, in terms of STD, SR and $\hat{\alpha}$. The standard deviations of SA and SB are statistically different from the ones provided by EW in the majority of cases.

To summarize, the analyses described above highlight the improvements provided by our combining methods when compared to well-known benchmarks in the literature, under both a statistical and a financial viewpoint. The results highlight the capability of our combinations in capturing the synergies of different methods designed to quantify different sources of causality in quantiles. Among them, we highlight the contribution of QN, that builds on the nonparametric test proposed by Jeong et al. (2012).

⁹ The network sparsity does not allow computing the eigenvector centrality with the power method (as explained in Appendix E) for the corresponding strategies SA-GR and SB-GR when using the 25 insurance companies dataset in Table 9 as in Appendix J.

Table 9

The table reports the performance of the strategies listed in the first column: (i) SA, that is long on stocks with low centrality and short on those with high centrality; (ii) SB, that is, the opposite of SA; the equally weighted portfolio (EW). The stocks' centrality is retrieved from five different networks: (i) Q(10%) that combines QB, Qo, QN (estimated for $\tau = 0.1$) and GR; (ii) Q(10%)\QN, that combines QB, Qo and GR; (iii) Q(10%)\Qo, that combines QB, QN and GR; (iv) Q(10%)\QB, that combines Qo, QN and GR; (v) GR, that builds on the Granger causality only. The second column indicates whether a common factor has been used (Y) or not (N) when estimating the underlying network. From left to right we report the following performance measures: the standard deviation (STD, %), the Sharpe ratio (SR, %; for both STD and SR, ***, ** and * indicate whether the performance of a given strategy is statistically different from the one generated by EW at the 1%, 5% and 10% significance level, respectively, by implementing the (Ledoit and Wolf, 2008)'s tests) and the Jensen's α ($\hat{\alpha}$; here, ***, ** and * indicate whether $\hat{\alpha}$ is statistically significant at the 1%, 5% and 10% level, respectively).

Strategy	Factor	Period 2006–2008			Period 2011–2015		
		STD	SR	$\hat{\alpha}$	STD	SR	$\hat{\alpha}$
(a) 25 Banks							
SA–Q(10%)	N	3.229***	0.091**	0.300	1.460***	–0.064	–0.047
SA–Q(10%)\QN	N	3.903	0.008	0.001	1.460***	–0.064	–0.047
SA–Q(10%)\Qo	N	2.512***	0.051**	0.077	1.452***	–0.010	0.006
SA–Q(10%)\QB	N	4.220	0.053	0.194	1.460***	–0.064	–0.047
SA–Q(10%)	Y	3.937	0.025	0.090	1.347***	–0.099	–0.098
SA–Q(10%)\QN	Y	3.219***	0.109**	0.302	1.344***	–0.100	–0.113
SA–Q(10%)\Qo	Y	2.971***	–0.101	–0.304	1.236***	0.056	0.048
SA–Q(10%)\QB	Y	3.937	0.025	0.090	1.347***	–0.099	–0.098
SB–Q(10%)	N	3.229***	–0.137	–0.452*	1.460***	0.063	0.046
SB–Q(10%)\QN	N	3.903	–0.046	–0.153	1.460***	0.063	0.046
SB–Q(10%)\Qo	N	2.512***	–0.111	–0.229	1.452***	0.009	–0.007
SB–Q(10%)\QB	N	4.220	–0.089	–0.346	1.460***	0.063	0.046
SB–Q(10%)	Y	3.937	–0.063	–0.242	1.347***	0.098	0.096
SB–Q(10%)\QN	Y	3.219***	–0.156	–0.453**	1.344***	0.099	0.111
SB–Q(10%)\Qo	Y	2.971***	0.050**	0.153	1.236***	–0.057	–0.050
SB–Q(10%)\QB	Y	3.937	–0.063	–0.242	1.347***	0.098	0.096
SA–GR	N	3.451**	–0.055	–0.271	1.103***	0.046	0.061
SA–GR	Y	2.963***	–0.288**	–0.827***	1.350***	0.053	0.108
SB–GR	N	3.451**	0.011	0.119	1.103***	–0.048	–0.062
SB–GR	Y	2.963***	0.237***	0.676***	1.350***	–0.054	–0.109
EW	–	4.714	–0.108	–0.333**	2.651	0.037	–0.070
(b) 25 Insurance companies							
SA–Q(10%)	N	4.118**	–0.105	–0.446	1.572***	–0.060	–0.061
SA–Q(10%)\QN	N	6.003*	0.049	0.176	1.572***	–0.060	–0.061
SA–Q(10%)\Qo	N	3.556**	–0.119	–0.453	1.719***	–0.070**	–0.157
SA–Q(10%)\QB	N	3.714**	–0.079	–0.337	1.565***	–0.088**	–0.105
SA–Q(10%)	Y	4.268	–0.155	–0.627*	1.543***	–0.081**	–0.110
SA–Q(10%)\QN	Y	5.053	0.011	–0.005	1.542***	–0.100**	–0.126
SA–Q(10%)\Qo	Y	3.556**	–0.119	–0.453	1.689**	–0.097**	–0.168
SA–Q(10%)\QB	Y	4.268	–0.155	–0.627*	1.543***	–0.081**	–0.110
SB–Q(10%)	N	4.118**	0.069	0.294	1.572***	0.059	0.060
SB–Q(10%)\QN	N	6.003*	–0.074	–0.328	1.572***	0.059	0.060
SB–Q(10%)\Qo	N	3.556**	0.076*	0.301	1.719***	0.069	0.155
SB–Q(10%)\QB	N	3.714**	0.039	0.186	1.565***	0.087	0.103
SB–Q(10%)	Y	4.268	0.120*	0.476	1.543***	0.080	0.109
SB–Q(10%)\QN	Y	5.053	–0.041	–0.147	1.542***	0.099	0.124
SB–Q(10%)\Qo	Y	3.556**	0.076*	0.301	1.689**	0.096	0.166
SB–Q(10%)\QB	Y	4.268	0.120*	0.476	1.543***	0.080	0.109
SA–GR	N	4.212***	0.089*	0.276	–	–	–
SA–GR	Y	4.723	–0.056	–0.293	–	–	–
SB–GR	N	4.212***	–0.124	–0.427	–	–	–
SB–GR	Y	4.723	0.024**	0.141	–	–	–
EW	–	5.459	–0.117	–0.391**	2.379	0.066	–0.046
(c) 48 Industry portfolios							
SA–Q(10%)	N	1.605***	0.024	–0.006	1.287***	–0.202***	–0.209***
SA–Q(10%)\QN	N	1.590***	0.028	0.001	1.287***	–0.202***	–0.209***
SA–Q(10%)\Qo	N	1.851***	–0.068	–0.188	1.165***	–0.148***	–0.174***
SA–Q(10%)\QB	N	1.914***	–0.099	–0.237	1.319***	–0.214***	–0.232***
SA–Q(10%)	Y	1.512***	0.054	0.055	1.147***	–0.156**	–0.107*
SA–Q(10%)\QN	Y	1.512***	0.054	0.055	1.177***	–0.135**	–0.077

(continued on next page)

3.4. Robustness checks

The results discussed in Section 3.3 focus on the combination of the four competing networks. In Section 3.3, we use weekly returns and allow for a heterogeneous impact of the composite network on the assets returns. In this section, we perform the

Table 9 (continued).

Strategy	Factor	Period 2006–2008			Period 2011–2015		
		STD	SR	$\hat{\alpha}$	STD	SR	$\hat{\alpha}$
SA-Q(10%)\Qo	Y	1.581***	−0.110	−0.223*	0.923***	0.062	0.028
SA-Q(10%)\QB	Y	1.394***	−0.039	−0.085	1.179***	−0.154**	−0.118*
SB-Q(10%)	N	1.605***	−0.118	−0.145	1.287***	0.201	0.208***
SB-Q(10%)\QN	N	1.590***	−0.122	−0.152	1.287***	0.201	0.208***
SB-Q(10%)\Qo	N	1.851***	−0.013	0.037	1.165***	0.147	0.172***
SB-Q(10%)\QB	N	1.914***	0.020	0.086	1.319***	0.213	0.230***
SB-Q(10%)	Y	1.512***	−0.153	−0.206*	1.147***	0.155	0.106
SB-Q(10%)\QN	Y	1.512***	−0.153	−0.206*	1.177***	0.133	0.076
SB-Q(10%)\Qo	Y	1.581***	0.015	0.071	0.923***	−0.064	−0.029
SB-Q(10%)\QB	Y	1.394***	−0.069	−0.067	1.179***	0.153	0.117*
SA-GR	N	1.896***	−0.117	−0.261*	0.879***	−0.079*	−0.018
SA-GR	Y	1.691***	−0.138	−0.284**	0.699***	−0.075*	−0.017
SB-GR	N	1.896***	0.038	0.109	0.879***	0.077	0.017
SB-GR	Y	1.691***	0.049	0.132	0.699***	0.073	0.016
EW	–	3.422	−0.053	0.052	2.194	0.090	−0.038

following robustness checks: (i) we combine only three networks; (ii) we estimate the model using monthly data; and (iii) we control for the optimal combination when the network impact is homogeneous across assets.

Appendix K reports tables assessing the model-based network combination when we exclude a network. We consider two specific cases. The first excludes the network estimated from Granger causality, in order to highlight the relevance of competing quantile causality networks. The second disregards the less relevant quantile causality network (the so-called baseline quantile causality). All the analyses are based on weekly data. The results confirm the relevance of the nonparametric approach for the purpose of obtaining a quantile causality that, in several cases, generates the largest and statistically most significant coefficients. The baseline quantile causality is the least relevant, even when the Granger causality network is excluded from the layers. The results for the industry portfolios are the most heterogeneous in terms of network relevance and they show a little improvement in terms of contraction of the residual correlations. For the banks and insurance companies, there is a marked gain in moving from linear factor models, or Granger causality augmented factor models, to a model that accounts for the presence of quantile causality; indeed, the average residual correlation decreases considerably.

Appendix G reports estimates of the network combination modelled on monthly data (as opposed to the weekly data considered in the previous section).¹⁰ We confirm the importance of nonparametric quantile causality and the limited impact of Granger causality. Once again, the outcome is clearer for the banks and insurances datasets. In addition, in contrast to the evidence emerging from the weekly data, the contraction of the residual correlations for the insurance company dataset is less evident during the financial crisis, whereas it is striking in the second period. Given the heterogeneity of the network impact on returns, as mentioned at the end of the previous section, we study the model performance when we impose a common reaction of the returns to exposure to the network. Appendix L illustrates the outcomes for the composite network and the impact of the network on returns in the three datasets and two sample periods, when the model is estimated on weekly data. The only relevant changes concern the composite network parameters: we now find a larger number of statistically significant coefficients. In addition, Granger causality seems to become more relevant, in the insurance companies dataset at least. The nonparametric quantile causality network nonetheless retains its role and receives the largest coefficients. Moving to the model-based combination confirms the previous results and enables a reduction in the average residual correlations in the banks and insurance companies datasets.

4. Concluding remarks

Causality networks have attracted a relevant attention in the financial economics literature during the recent years for the purposes of systemic risk interpretation. We show that the structure of causality networks changes considerably if we move from the mean – commonly associated to the standard Granger causality – to the extreme quantiles. In the latter case, the causality across assets can be measured by means of quantile regression-based approaches. Here we illustrate three different possibilities: the traditional quantile regression introduced by [Koenker and Bassett \(1978\)](#), its generalization to the quantile-on-quantile approach of [Sim and Zhou \(2015\)](#) and the nonparametric test due to [Jeong et al. \(2012\)](#). From a systemic risk perspective, we show that moving to quantile-based approaches leads to denser networks with a broader systemic relevance. By focusing on a linear factor model augmented with a multi-layer network dependence, we also demonstrate that causality networks are useful for capturing this dependence across financial returns, going beyond what can be explained by a market factor. This paves the way to further analyses and has an associated impact on diversification analyses and portfolio construction, as well as on asset pricing.

Appendix A. Supplementary data

Supplementary material related to this article containing all the Appendices mentioned in the text can be found online at <https://doi.org/10.1016/j.jempfin.2019.08.008>.

¹⁰ In this case too, we ran a sensitivity analysis on the effects of excluding a network. Tables reporting these results are available upon request.

References

- Acemoglu, D., Carvalho, D.M., Ozdaglar, A., 2012. The network origins of aggregate fluctuations. *Econometrica* 80 (5), 1977–2016.
- Anselin, L., 1988. *Spatial Econometrics: Methods and Models*. Springer, Netherlands.
- Balcilar, M., Bekiros, S., Gupta, R., 2017. The role of news-based uncertainty indices in predicting oil markets: a hybrid nonparametric quantile causality method. *Empir. Econom.* 53, 879–889.
- Bargigli, L., di Iasio, G., Infante, L., Lillo, F., Pierobon, F., 2015. The multiplex structure of interbank networks. *Quant. Finance* 15 (4), 673–691.
- Barigozzi, M., Brownlee, C., 2019. NETS: Network estimation for time series. *J. Appl. Econometrics* 34 (3), 347–364.
- Billio, M., Caporin, M., Frattarolo, L., Pelizzon, L., 2016. Networks in risk spillovers: a multivariate GARCH perspective, University Ca' Foscari of Venice, Dept. of Economics Research Paper Series No. No. 03/WP/ 2016.
- Billio, M., Caporin, M., Panzica, R., Pelizzon, L., 2017. The impact of network connectivity on factor exposures, asset pricing and portfolio diversification, SAFE Working Paper No. 166.
- Billio, M., Getmansky, M., Lo, A.W., Pelizzon, L., 2012. Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *J. Financ. Econ.* 104 (3), 535–559.
- Billio, M., Gray, D., Getmansky, M., Lo, A., Merton, R., Pelizzon, L., 2014. Sovereign, bank and insurance credit spreads: connectedness and system networks, MIT Working Paper.
- Bonaccolto, G., Caporin, M., Gupta, R., 2018. The dynamic impact of uncertainty in causing and forecasting the distribution of oil returns and risk. *Physica A* 507, 446–469.
- Bowman, A., 1984. An alternative method of cross-validation for the smoothing of kernel density estimates. *Biometrika* 71, 353–360.
- Brandsma, A.S., Ketellapper, R.H., 1979. A biparametric approach to spatial autocorrelation. *Environ. Plan. A* 11, 51–58.
- Candelon, B., Topkavi, S., 2016. A nonparametric test for granger causality in distribution with application to financial contagion. *J. Bus. Econom. Statist.* 34 (2), 240–253.
- Caporin, M., Paruolo, P., 2015. Proximity-structured multivariate volatility models. *Econometric Rev.* 34 (5), 559–593.
- Carhart, M., 1997. On the persistence of mutual fund performance. *J. Finance* 52, 57–82.
- Cheung, Y.W., Ng, L.K., 1996. A causality-in-variance test and its application to financial market prices. *J. Econometrics* 72, 33–48.
- Comte, F., Lieberman, O., 2000. Second-order noncausality in multivariate GARCH processes. *J. Time Series Anal.* 21 (5), 535–557.
- Cont, R., 2001. Empirical properties of asset returns: stylized facts and statistical issues. *Quant. Finance* 1, 223–236.
- Corsi, F., Lillo, F., Pirino, D., 2015. Measuring flight-to-quality with granger-causality tail risk networks. Available at SSRN 2576078.
- DeMiguel, V., Garlappi, L., Uppal, R., 2009. Optimal versus naive diversification: how inefficient is the 1/N portfolio strategy? *Rev. Financ. Stud.* 22, 1915–1953.
- Diebold, F.X., Yilmaz, K., 2014. On the network topology of variance decompositions: measuring the connectedness of financial firms. *J. Econometrics* 182 (1), 119–134.
- Diebold, F.X., Yilmaz, K., 2015. *Financial and Macroeconomic Connectedness*. Oxford University Press.
- Duchin, R., Levy, H., 2009. Markowitz versus the talmudic portfolio diversification strategies. *J. Portf. Manag.* 35, 71–74.
- Efron, B., Tibshirani, R., 1993. *An Introduction to the Bootstrap*. Chapman & Hall, New York.
- Elhorst, J.P., Lacombe, D.J., Piras, G., 2012. On model specification and parameter space definitions in higher order spatial econometric models. *Reg. Sci. Urban Econ.* 42 (1), 211–220.
- Fama, E., French, K., 1993. Common risk factors in the returns of stocks and bonds. *J. Financ. Econ.* 33 (3–53), 3–56.
- Fama, E., French, K., 1995. Size and book-to-market factors in earnings and returns. *J. Finance* 50 (1), 131–155.
- Fruchterman, T.M., Reingold, E.M., 1991. Graph drawing by force-directed placement. *Softw. - Pract. Exp.* 21 (11), 1129–1164.
- Geweke, J., 1984. Inference and causality in economic time series. In: *Handbook of Econometrics*, Vol. 2. pp. 1101–1144.
- Granger, C.W.J., 1969. Investigating causal relations by econometric models and cross spectral methods. *Econometrica* 37, 424–438.
- Granger, C.W.J., 1980. Testing for causality: a personal viewpoint. *J. Econ. Dyn. Control* 2 (4), 329–352.
- Granger, C.W.J., 1988. Some recent developments in a concept of causality. *J. Econometrics* 39, 199–211.
- Granger, C.W.J., Robins, R.P., Engle, R.F., 1986. Wholesale retail prices: bivariate time series modeling with forecastable error variances. In: *Model Reliability*. MIT Press, Boston MA, pp. 1–17.
- Hafner, C.M., 2003. Fourth moments of multivariate GARCH models. *J. Financ. Econ.* 1 (1), 26–54.
- Hafner, C.M., Herwartz, H., 2008. Testing for causality in variance using multivariate GARCH models. *Ann. Écon. Stat.* 89, 215–241.
- Hautsch, N., Kyj, L.M., Malec, P., 2013. Do High-Frequency Data Improve Highdimensional Portfolio Allocations? Technical Report, Sonderforschungsbereich 649, Humboldt University, Berlin, Germany.
- Hautsch, N., Kyj, L.M., Oomen, R.C., 2012. A blocking and regularization approach to high-dimensional realized covariance estimation. *J. Appl. Econometrics* 27 (4), 625–645.
- Hautsch, N., Schaumburg, J., Schienle, M., 2014. Forecasting systemic impact in financial networks. *Int. J. Forecast.* 30 (3), 781–794.
- Hautsch, N., Schaumburg, J., Schienle, M., 2015. Financial network systemic risk contributions. *Rev. Finance* 19 (2), 685–738.
- Hong, Y., Liu, Y., S, W., 2009. Granger causality in risk and detection of extreme risk spillover between financial markets. *J. Econometrics* 150, 271–287.
- Hoover, K.D., 2001. *Causality in Macroeconomics*. Cambridge University Press.
- Jaccard, P., 1901. Étude comparative de la distribution florale dans une portion des alpes et des jura. *Bull. Soc. Vaud. Sci. Nat.* 37, 547–579.
- Jeong, K., Hardle, W.K., Song, S., 2012. A consistent nonparametric test for causality in quantile. *Econometric Theory* 28, 861–887.
- Kelly, B., Lustig, H., Nieuwerburgh, S.V., 2013. Firm volatility in granular networks, mimeo. Booth School of Business, UCLA-Anderson and Stern School of Business.
- Koenker, R., Bassett, G., 1978. Regression quantiles. *Econometrica* 46 (1), 33–50.
- Ledoit, O., Wolf, M., 2008. Robust performance hypothesis testing with the Sharpe ratio. *J. Empir. Financ.* 15, 850–859.
- Ledoit, O., Wolf, M., 2011. Robust performance hypothesis testing with the variance. *Willmott Mag.* 55, 86–89.
- Lee, L.F., Liu, X., 2010. Efficient GMM estimation of higher order spatial autoregressive models with autoregressive disturbances. *Econometric Theory* 26, 187–230.
- Lee, T., Yang, W., 2014. Granger-causality in quantiles between financial markets: Using copula approach. *Int. Rev. Financ. Anal.* 33, 70–78.
- Léon, C., Berndsen, R., Renneboog, L., 2014. Financial stability and interacting networks of financial institutions and market infrastructures, center discussionpaper Series, 69.
- LeSage, J., Pace, R.K., 2009. *Introduction to Spatial Econometrics*. Chapman and Hall/CRC, Taylor and Francis Group, Boca Raton, FL, USA.
- Molina-Balboa, J., Martínez-Jaramillo, S., van der Leij, M., Lopez-Gallo, F., 2014. A multiplex network analysis of the mexican banking system: link persistence, overlap and waiting times. *J. Netw. Theory Finance* 1 (1), 99–138.
- Montagna, M., Kok, C., 2013. Multi-layered interbank model for assessing systemic risk, kiel Institute for the World Economy Working Paper No. 1873 and ECB Working Paper.
- Ozdagli, A., Weber, M., 2017. Monetary Policy Through Production Networks: Evidence from the Stock Market. Tech. Rep., National Bureau of Economic Research.
- Rudemo, M., 1982. Empirical choice of histograms and kernel density estimators. *Scand. J. Stat.* 9, 65–78.
- Sim, N., Zhou, H., 2015. Oil prices, US stock return, and the dependence between their quantiles. *J. Bank. Financ.* 55, 1–8.
- Tu, J., Zhou, G., 2011. Markowitz meets Talmud: A combination of sophisticated and naive diversification strategies. *J. Financ. Econ.* 99 (1), 204–215.